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Intelligent System for Predicting Umrah Crowd Levels and Optimal Visiting Times at Al-Masjid Al-Haram.

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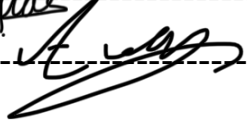
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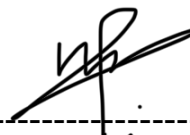
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Table of Contents

<i>Chapter 1 Introduction</i>	<i>7</i>
1.1 Project Motivation	7
1.2 Problem Statement	7
1.3 Project Contribution.....	9
1.4 Structure of the Project	9
<i>Chapter 2 : Related Work</i>	<i>11</i>
2.1 Crowd Management in Religious Environments.....	11
2.2 Intelligent Forecasting and Crowd Analytics for Umrah Crowd	14
<i>2.2.1 Predictive Analytics and Machine Learning for Crowd Behavior</i>	<i>15</i>
<i>2.2.2 Umrah and Religious Visitor Flow Analytics.....</i>	<i>16</i>
2.3 Limitations of Previous Studies and Research Gap Summary	17
<i>2.3.1 Limitations of Previous Studies</i>	<i>17</i>
<i>2.3.2 Research Gaps</i>	<i>18</i>
2.4 Summary.....	19
<i>Chapter 3 Data Collection and Preparation</i>	<i>20</i>
3.1 Introduction	20
3.2 Data Sources	21
3.3 Raw Data Description	23
3.4 Data Structure and Challenges	25
3.5 Data Integration.....	26
3.6 Data Cleaning	27
3.7 Data Transformation	29
3.8 Time Series Preparation	30
3.9 Synthetic Data Generation.....	32

3.9.1 Daily Distribution.....	32
3.9.2 Weighted Distribution.....	33
3.9.3 Ramadan Handling.....	33
3.9.4 Dhul-Qi'dah 1446H Estimation.....	34
3.9.5 Hajj Feature Construction	34
3.9.6 Assumptions.....	35
3.10 Feature Engineering.....	35
3.11 Crowding Level Construction	37
3.12 Final Dataset.....	39
3.13 Limitations of the Data.....	40
3.14 Chapter Summary	41
<i>Chapter 4: Methodology and Implementation</i>	<i>43</i>
4.1 Overview of the Methodology and Implementation	43
4.2 Overall System Architecture	44
4.3 Data Flow and Processing Pipeline	46
4.4 Forecasting Model Development	48
4.5 Feature Engineering and Time-Series Modeling	50
4.6 Interactive Dashboard and Recommendation Implementation.....	52
4.7 System Implementation Environment	54
4.8 Chapter Summary	55
<i>Chapter 5: Results and Discussion.....</i>	<i>57</i>
5.1 Introduction	57
5.2 Experimental Results.....	57
5.2.1 Baseline and Initial Model Results	58
5.2.2 Hybrid Model Results.....	58
5.2.3 Final Adopted Forecasting Model	60
5.3 Ramadan and Seasonal Analysis.....	62
5.4 Dashboard Results and Recommendation Outputs	64
5.4.1 Main Dashboard Interface.....	65

5.4.2 User Input Page.....	66
5.4.3 Non-Saudi Restriction During Dhul-Qi'dah	67
5.4.4 Selected-Day Forecasting Output Cards	68
5.4.5 Final Recommendation Output	68
5.4.6 Seasonal Indicators for Dhul-Qi'dah and Dhul-Hijjah.....	69
5.4.7 Seven-Day Forecast Visualization	70
5.4.8 Best-Day Recommendation Component	71
5.4.9 Dashboard Result Summary	71
5.5 Discussion.....	72
5.5.1 Why Growth-Rate XGBoost Performed Better	72
5.5.2 Why Ramadan Was More Difficult to Forecast.....	74
5.5.3 Interpretation of the Hybrid Forecasting Experiments	75
5.5.4 Strengths and Weaknesses of the Proposed System	76
5.6 Limitations.....	77
5.7 Chapter Summary	78
Chapter 6: Conclusion and Future Work	80
6.1 Introduction	80
6.2 Project Overview and Achievements	80
6.3 Main Findings	81
6.4 Limitations of the Current Work.....	81
6.5 Project Conclusion	82
6.6 Future Work.....	82
6.7 Final Remarks	83
References	84
Appendix	87

List of Tables

Table 3.1 Description of Raw Datasets Before Preprocessing	24
Table 3.2 Description of Final Dataset Columns After Preprocessing	28
Table 3.3 Final Dataset Columns	41
Table 3.4 Example of the Final Dataset Structure	42
Table 4.1 Main Components of the Proposed Methodology	49
Table 4.2 Overall Architecture Layers of the Proposed System	51
Table 4.3 Tools and Technologies Used in the System Implementation	56
Table 5.1 Baseline and Initial Forecasting Model Results	60
Table 5.2 Hybrid Forecasting Model Results	61
Table 5.3 Final Adopted Forecasting Model Performance	63
Table 5.4 Ramadan and Seasonal Forecasting Analysis	65
Table 5.5 Dashboard Output Summary	74

List of Figures

Figure 3.1 Data Preprocessing Pipeline for Preparing the Final Forecasting Dataset	20
Figure 4.1 Overall System Architecture of the Umrah Forecasting System	46
Figure 4.2 Data Flow and Processing Pipeline	48
Figure 5.1 Home Page of the Umrah Visitors Smart Forecasting System Dashboard	67
Figure 5.2 Dashboard Input Page	68
Figure 5.3 Non-Saudi Restriction Message During Dhul-Qi'dah	69
Figure 5.4 Selected-Day Forecasting Output Cards	70
Figure 5.5 Final Recommendation Output	71
Figure 5.6 Seasonal Indicators for Dhul-Qi'dah and Dhul-Hijjah	71
Figure 5.7 Seven-Day Umrah Crowd Forecast Visualization	72
Figure 5.8 Optional Best-Day Recommendation Button	73
Figure 5.9 Best-Day Recommendation Output	73

Abstract

Crowd forecasting and decision-support systems are important areas in data science, especially for high-density religious environments where crowd levels affect comfort, safety, and visit planning. Al-Masjid Al-Haram receives millions of Umrah performers throughout the year, and crowd density changes significantly across normal days, weekends, Ramadan, and Hajj-related periods. This makes it difficult for Umrah performers to predict congestion before arrival and select suitable visiting days.

This project proposes an intelligent forecasting and recommendation system for predicting daily Umrah performer counts and expected crowd levels at Al-Masjid Al-Haram. The system uses prepared historical Umrah-related data, time-series feature engineering, and machine learning models to identify crowd patterns and forecast future crowd behavior. Several forecasting approaches were evaluated, and the Standalone Growth-Rate XGBoost model was adopted as the final model due to its stronger performance and suitability for dashboard integration.

The final system presents the results through an interactive Arabic dashboard that displays predicted Umrah counts, crowding levels, temperature, seasonal indicators, seven-day forecasts, and best-day recommendations. The system supports Umrah performers in planning visits more effectively, reducing uncertainty, and improving the overall Umrah experience.

Chapter 1 Introduction

1.1 Project Motivation

Crowd management in large religious environments has become an important challenge due to the high density of people, limited spatial capacity, and the synchronized nature of religious rituals. In Al-Masjid Al-Haram, this challenge becomes more critical because large numbers of Umrah performers move through specific ritual areas such as Tawaf around the Kaaba and Sa'i between Safa and Marwah.

The number of Umrah performers does not remain constant throughout the year. It changes according to Hijri months, weekdays, religious seasons, holidays, Ramadan, and Hajj-related periods. These variations can lead to sudden increases in crowd levels, making it difficult for Umrah performers to choose suitable times for performing their rituals comfortably and safely.

Many existing crowd management approaches focus mainly on monitoring current crowd conditions or supporting operational authorities. However, Umrah performers also need simple and understandable predictive information before they arrive. They need to know the expected crowd level and whether there is a better day to perform Umrah within the near future.

Therefore, this project is motivated by the need to develop an intelligent forecasting system that predicts expected Umrah crowd levels at Al-Masjid Al-Haram. The system uses historical Umrah-related data and machine-learning techniques to provide useful predictions. The final output is presented through an interactive dashboard that displays the expected number of Umrah performers, classifies the crowding level, and suggests the best available visiting day within the next seven days when needed.

1.2 Problem Statement

Although crowd monitoring and management systems have improved in Al-Masjid Al-Haram, many existing approaches remain reactive rather than

predictive. They mainly support the observation and control of current crowd conditions, but they do not always provide forward-looking information that can help Umrah performers plan their visit before arrival.

At present, Umrah performers may not have access to a simple predictive tool that estimates future crowd levels based on historical Umrah patterns, seasonal behavior, and temporal features. This creates uncertainty when selecting suitable times to perform Umrah rituals, especially during high-demand periods such as Ramadan, weekends, holidays, and Hajj-related periods. As a result, some Umrah performers may arrive during crowded periods without prior awareness, which can increase waiting time, reduce comfort, and make ritual performance more difficult.

Previous research has shown that forecasting methods can support early congestion identification and proactive planning in high-density environments [1], [2]. However, many existing studies are either designed for operational decision-makers or focus on crowd monitoring and control rather than direct guidance for individual Umrah performers [1], [8]. This creates a gap between crowd-analysis research and the practical needs of Umrah performers who require simple, understandable, and timely recommendations.

The central problem addressed in this project is the lack of a data-driven, Umrah-focused forecasting system that predicts expected crowd levels for Umrah performers at Al-Masjid Al-Haram and recommends suitable visiting times. The project aims to bridge this gap by analyzing historical Umrah patterns, extracting temporal and seasonal features, applying forecasting techniques, and presenting the results through an accessible dashboard.

1.3 Project Contribution

This project contributes to the field of Umrah crowd forecasting and decision-support systems by developing a data-driven forecasting system focused specifically on Umrah performers at Al-Masjid Al-Haram.

First, the project introduces an Umrah-focused predictive approach that uses historical Umrah-related data to estimate future crowd levels. Instead of treating the problem as general visitor forecasting, the project focuses on Umrah performers and their seasonal and ritual-related patterns.

Second, the project supports proactive planning by shifting the focus from only observing current crowd conditions to predicting future crowd levels. This helps Umrah performers make better decisions before arriving at Al-Masjid Al-Haram.

Third, the project applies machine-learning forecasting techniques to predict the expected number of Umrah performers. These predictions are then transformed into understandable crowd-level categories, making the results easier for non-technical users to understand.

Fourth, the project develops an interactive dashboard that presents the predicted number of Umrah performers, the expected crowd level, and a practical recommendation. A key contribution of the dashboard is that it can suggest the best available day within the next seven days, allowing Umrah performers to compare nearby days and choose a more suitable time for their visit.

Finally, the project connects forecasting results with user-oriented recommendations. This makes the system not only an analytical forecasting tool, but also a practical decision-support tool for Umrah performers.

1.4 Structure of the Project

This report is organized into six chapters.

Chapter One introduces the project motivation, problem statement, project contribution, and the overall structure of the report.

Chapter Two reviews previous studies related to crowd management, religious mass gatherings, Umrah congestion forecasting, and data-driven forecasting approaches. It also explains how the current project builds on previous work by focusing on Umrah performers and dashboard-based recommendations.

Chapter Three presents the data collection and preprocessing process, including data sources, raw data description, data cleaning, data integration, time-series preparation, synthetic daily data generation, feature engineering, and final dataset construction.

Chapter Four presents the methodology and implementation of the proposed system, including the overall system architecture, forecasting pipeline, model development process, dashboard implementation, recommendation logic, and system implementation environment.

Chapter Five presents the results and discussion, including experimental results, baseline model comparison, hybrid model analysis, Ramadan and seasonal analysis, dashboard outputs, recommendation examples, limitations, and interpretation of the forecasting results.

Chapter Six concludes the project by summarizing the main achievements and findings, discussing the project limitations, and suggesting future improvements such as using real daily data, real-time crowd data, sensors, personalized recommendations, and AI-based chatbot support.

Chapter 2 : Related Work

This chapter reviews previous studies related to crowd management, religious mass gatherings, Umrah congestion forecasting, and data-driven predictive analytics. The chapter focuses on studies that are relevant to the project's aim of predicting crowd levels for Umrah performers at Al-Masjid Al-Haram. It begins by discussing crowd management in religious environments, then reviews intelligent forecasting and crowd analytics approaches. Finally, it summarizes the limitations of previous studies and identifies the main research gaps addressed by this project.

2.1 Crowd Management in Religious Environments

Crowd management refers to the planning, monitoring, and organization of large groups of people in shared spaces to maintain safety, comfort, and smooth movement. In high-density environments, poor crowd management can lead to congestion, reduced walking speed, pressure accumulation, and increased safety risks. These challenges become more significant in religious environments because crowd movement is often linked to fixed rituals, specific locations, and synchronized timing [4], [7].

In the context of Umrah, crowd management is especially important because Umrah performers follow specific ritual paths, including Tawaf around the Kaaba and Sa'i between Safa and Marwah. These rituals are performed in defined spaces inside Al-Masjid Al-Haram, where movement capacity is limited and congestion can develop quickly during peak periods. Therefore, understanding and predicting the movement of Umrah performers is essential for improving comfort, supporting safer movement, and reducing crowd-related difficulties.

Previous research on crowd dynamics has shown that high-density religious sites require structured planning and proactive coordination. Studies conducted in

religious environments emphasize that variations in crowd volume strongly affect movement flow and safety [4], [7]. These studies support the need for systems that can analyze crowd behavior, anticipate congestion, and provide timely guidance.

2.1.1 Crowd Management as a Global Challenge in Religious Events

Religious mass gatherings such as Hajj and Umrah are among the most complex crowd environments in the world. Unlike general public events, religious rituals are performed in fixed locations and often require large numbers of people to move in similar directions within limited time periods. This creates synchronized movement patterns that can increase congestion risks [7].

Owaidah et al. studied crowd behavior during Tawaf and Sa'i using discrete-event simulation and agent-based modeling. Their work showed that operational and spatial factors, such as pathway width, gate allocation, and crowd density, can significantly affect the time required to complete rituals [7]. This indicates that crowd conditions in religious rituals are sensitive to both space and movement flow.

Derhab et al. proposed a crowd congestion forecasting framework for the Umrah use case using ensemble learning and decision-making techniques. Their study demonstrated that data-driven forecasting can help identify congestion before it occurs [1]. This is directly related to the current project because both focus on Umrah and the use of predictive analytics to support proactive crowd awareness. Alharthi et al. also examined the use of data mining techniques for Hajj crowd management. Their study highlighted the importance of intelligent systems in identifying crowd risks and supporting safer movement in pilgrimage environments [6]. Although their focus was more related to Hajj, the findings are relevant because Hajj and Umrah share similar crowd-management challenges, including high density, ritual movement, and limited spatial flexibility.

Together, these studies show that religious crowd management requires more than real-time observation. It requires predictive and intelligent systems capable of anticipating congestion and supporting proactive planning.

2.1.2 Operational Challenges of Umrah Crowd Management

Umrah crowd management has several operational challenges that distinguish it from general crowd management. First, Umrah performers move through fixed ritual zones, especially the Mataf area and the Sa'i corridor. These areas have specific spatial layouts, which limits the ability to redistribute crowds when congestion increases [7].

Second, Umrah demand varies strongly across time. Ramadan is one of the most important peak periods, as the number of Umrah performers often increases sharply. Weekends, holidays, and Hajj-related periods can also affect the number of people performing Umrah. These temporal patterns make forecasting important because crowd levels may change rapidly from one day to another.

Third, the Hijri calendar plays a major role in Umrah activity. Since religious behavior is strongly connected to Hijri months and Islamic events, forecasting systems should consider Hijri dates rather than relying only on Gregorian dates. This is especially important for identifying seasonal effects such as Ramadan and Dhul-Hijjah-related periods.

Fourth, the movement of Umrah performers is highly ritual-driven. Most Umrah performers follow a similar sequence of rituals, which may increase pressure on specific areas during peak periods. Previous studies on Tawaf and Sa'i confirm that ritual movement patterns and spatial constraints can strongly affect crowd flow and ritual completion time [7].

These operational challenges show that forecasting Umrah crowd levels requires a combination of historical Umrah patterns, temporal features, seasonal indicators, and machine-learning methods capable of capturing non-linear patterns.

2.1.3 Digital Transformation and the Role of Data in Crowd Systems

Digital transformation has improved the ability to collect, analyze, and use data for crowd management. Modern crowd systems may rely on mobility logs, sensing technologies, surveillance analytics, open data, and official statistical records to understand crowd behavior and support decision-making.

Yamada et al. demonstrated the use of multi-source mobility traces and transit-search logs for early crowd forecasting in transportation environments [2]. Although their study was not focused on Umrah, it shows the value of combining data sources to anticipate crowd conditions before congestion occurs.

In tourism and visitor-demand forecasting, Wu et al. used multi-source indicators and time-series analysis to understand visitation patterns and future demand [3]. Similarly, Xu et al. applied advanced time-series forecasting methods to predict visitor arrivals at tourist attractions [5]. These studies are relevant because they demonstrate how historical demand data can be transformed into predictive insights.

Within religious crowd environments, Derhab et al. demonstrated that historical Umrah-related data can be used in a forecasting framework to predict congestion states before they occur [1]. This supports the idea that data-driven forecasting can be applied to Umrah crowd prediction and transformed into practical decision-support outputs.

2.2 Intelligent Forecasting and Crowd Analytics for Umrah Crowd

Data-driven analytics have become important in crowd management because they allow systems to move from reactive monitoring toward predictive planning.

Instead of waiting for congestion to occur, forecasting models can analyze historical data and estimate future crowd levels. This direction is supported by studies that apply predictive analytics and machine learning to crowd and demand forecasting [1], [2], [3], [5].

In the context of Umrah, data-driven forecasting is particularly useful because crowd levels are influenced by recurring temporal and seasonal patterns. Historical Umrah records can show repeated increases during certain Hijri months, Ramadan, weekends, and religious periods. Machine-learning models can learn these patterns and generate predictions for future dates.

Derhab et al. are especially relevant to this project because their work focused on Umrah congestion forecasting using ensemble learning and decision-making methods [1]. Their study supports the idea that predictive analytics can be used to identify congestion states in advance. However, the current project differs by focusing more on transforming predictions into dashboard-based guidance for Umrah performers.

Other forecasting studies in transportation and tourism also support the project's methodology. They show that regression models, time-series decomposition, and advanced machine-learning techniques can be effective in predicting crowd or visitor demand when historical patterns are available [2], [3], [5]. These studies provide a broader foundation for using predictive analytics in the proposed Umrah forecasting system.

2.2.1 Predictive Analytics and Machine Learning for Crowd Behavior

Predictive analytics refers to the use of historical data and computational models to estimate future outcomes. In crowd management, predictive analytics can help estimate future density, identify peak periods, and support proactive planning.

Machine-learning models such as Random Forest, Gradient Boosting, and ensemble learning methods are widely used in forecasting and classification tasks because they can capture complex and non-linear relationships between input features and target values. Derhab et al. applied ensemble learning in the Umrah congestion forecasting context, which supports the suitability of such methods for religious crowd prediction [1]. Alharthi et al. also highlighted the role of data mining techniques in intelligent Hajj crowd management [6].

For Umrah forecasting, such models are suitable because the number of Umrah performers is affected by multiple interacting factors. These include previous

crowd levels, weekly patterns, Hijri months, Ramadan, Hajj-related days, and other contextual variables. Using engineered features allows the model to learn both short-term and seasonal changes in Umrah demand.

Classical and deep learning forecasting approaches also appear in the literature. For example, time-series forecasting approaches can capture general trends and seasonal cycles, while advanced models such as N-BEATS have been applied to visitor-arrival forecasting in tourism [5]. However, for this project, machine-learning models based on structured features are practical because they are compatible with the available data and suitable for integration into an interactive dashboard.

2.2.2 Umrah and Religious Visitor Flow Analytics

Visitor flow analytics in religious environments focuses on understanding how people move through ritual spaces and how congestion develops over time. In Al-Masjid Al-Haram, the most important movement areas include Tawaf, Sa'i, mosque entrances, exits, and surrounding access routes.

Owaidah et al. analyzed Tawaf and Sa'i rituals using simulation models. Their findings showed that spatial and operational factors influence crowd movement and ritual completion time [7]. However, simulation-based studies may not fully reflect actual daily and seasonal variations in Umrah demand.

Derhab et al. provided a more direct contribution to Umrah forecasting by using historical data and ensemble learning to predict congestion [1]. Their study demonstrated that Umrah crowd states can be forecasted before congestion occurs. This supports the feasibility of the current project's forecasting direction.

Alharthi et al. used machine-learning and data-mining methods for Hajj crowd management [6]. Although the study is focused on Hajj, it is useful for understanding how intelligent systems can support crowd-risk identification in pilgrimage settings.

Overall, previous studies show that both simulation-based and data-driven approaches are useful for religious crowd analysis. However, many of them are designed mainly for operational use and do not focus on providing simple forecasting guidance directly to Umrah performers. This limitation motivates the dashboard-oriented approach adopted in this project.

2.3 Limitations of Previous Studies and Research Gap Summary

2.3.1 Limitations of Previous Studies

Despite the value of previous studies, several limitations remain relevant to the current project.

First, some studies rely heavily on simulation-generated data. Simulation is useful for testing crowd scenarios, but it may not fully capture actual variations in Umrah demand across different days, months, and religious seasons [7].

Second, several studies focus on specific ritual locations such as Tawaf or Sa'i rather than forecasting the overall demand of Umrah performers over time. This limits their usefulness for general visit planning [7].

Third, many existing systems are designed mainly for operational authorities, such as crowd managers or decision-makers. For example, data-driven crowd decision-support systems are often developed to support planning and control at the management level rather than providing direct guidance for individual users [8].

Fourth, some studies emphasize real-time monitoring or congestion detection rather than forward-looking prediction. While real-time monitoring is important, it does not fully support advance planning for Umrah performers [1], [8].

Fifth, several forecasting studies outside the religious context demonstrate the value of time-series and multi-source data for predicting demand, but their direct application to Umrah performers remains limited [2], [3], [5]. This indicates the need for a system that adapts predictive analytics specifically to Umrah crowd patterns and user-oriented recommendations.

These limitations show the need for a system that focuses on Umrah performers, applies forecasting methods, and presents results in a simple dashboard designed for advance visit planning.

2.3.2 Research Gaps

Based on the reviewed studies, the following research gaps are identified:

Gap 1: Limited availability of Umrah-focused predictive tools that provide advance guidance for Umrah performers [1].

Gap 2: Many previous studies focus on operational crowd management rather than user-friendly recommendations for individual Umrah performers [6], [7], [8].

Gap 3: Simulation-based approaches are useful for analyzing ritual movement, but they may not fully capture real temporal and seasonal variations in Umrah demand [7].

Gap 4: Existing forecasting studies in transportation and tourism demonstrate strong predictive potential, but limited work adapts these approaches directly to Umrah crowd forecasting and dashboard-based guidance [2], [3], [5].

Gap 5: Limited dashboard-based systems translate numerical forecasts into clear crowd-level categories and suggested visiting times for Umrah performers.

The proposed project addresses these gaps by developing a data-driven forecasting system that predicts expected Umrah crowd levels and presents the results through an interactive dashboard designed to support Umrah performers in selecting suitable visiting times. Unlike previous studies that focused mainly on operational monitoring or simulation-based analysis, the proposed system focuses on transforming forecasting outputs into practical and user-oriented recommendation insights for Umrah performers.

2.4 Summary

This chapter reviewed previous studies related to crowd management, religious mass gatherings, Umrah congestion forecasting, and data-driven predictive analytics. The literature shows that religious environments such as Hajj and Umrah are highly complex because they involve large numbers of people, fixed ritual spaces, synchronized movement, and strong seasonal demand patterns [1], [6], [7].

The reviewed studies demonstrate the importance of simulation, machine learning, and predictive analytics in understanding and managing crowd behavior. Studies such as Owaidah et al. highlight the role of simulation in analyzing Tawaf and Sa'i movement [7], while Derhab et al. demonstrate the feasibility of forecasting Umrah congestion using data-driven methods [1]. Other studies in transportation and tourism forecasting further support the use of historical data and machine-learning models for predicting future demand [2], [3], [5].

However, the review also shows that many previous studies either rely on simulation data, focus on specific ritual zones, or target operational decision-makers rather than Umrah performers. In addition, limited work has translated forecasting outputs into simple dashboard-based recommendations for individual users.

Therefore, the proposed project addresses these gaps by focusing specifically on Umrah performers, applying forecasting techniques, and presenting the results through an interactive dashboard that provides expected crowd levels and suggested visiting times, including the best available day within a nearby seven-day period when needed.

Chapter 3 Data Collection and Preparation

3.1 Introduction

This chapter explains the data collection and preparation process used in this project. Since the project aims to predict Umrah performer counts and crowd levels at Al-Masjid Al-Haram, preparing a reliable daily dataset was an essential step.

The data was collected mainly from the Saudi Open Data Platform and included Umrah permits, Ramadan data, Hajj-related data, weather data, and Hijri/Gregorian calendar information. However, the original files had different formats, date systems, and time frequencies, so they required cleaning, transformation, integration, and feature engineering.

By the end of this stage, all data sources were combined into one daily dataset suitable for forecasting models, crowd-level analysis, and the dashboard recommendation system.

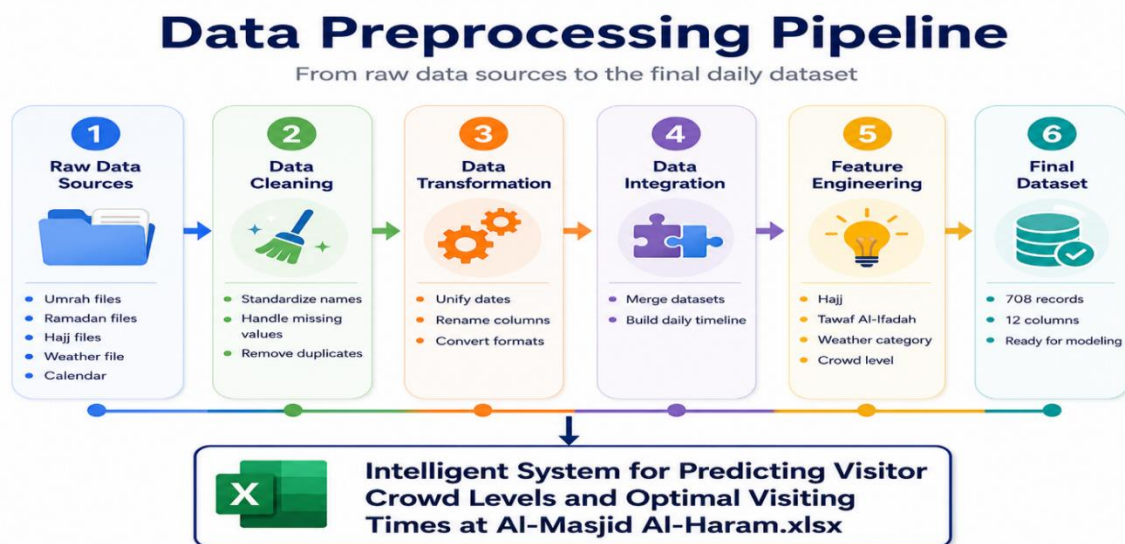


Figure 3.1: Data preprocessing pipeline for preparing the final forecasting dataset.

3.2 Data Sources

The data used in this project was obtained from the Saudi Open Data Platform, in addition to a Hijri/Gregorian calendar that was created during the preprocessing stage to link Hijri dates with Gregorian dates. Although most of the data came from official open sources, it was distributed across several files, with each file representing a different aspect of the factors affecting the number of Umrah performers and crowd levels at Al-Masjid Al-Haram. The Saudi Open Data Platform was selected because it provides official, reliable, and organized datasets. It was also suitable for this project because it includes data directly related to Umrah, Ramadan, Hajj, and weather, which are the main factors influencing visitor movement and crowding at Al-Masjid Al-Haram.

These datasets were selected because they are directly related to the objective of the project, which is to develop an intelligent system for predicting the number of Umrah performers, crowd levels, and suitable times for performing Umrah. Therefore, the project relied on data sources covering Umrah data, Ramadan data, Hajj data, weather data, and calendar data, as they represent the most important temporal, seasonal, and environmental factors affecting visitor movement.

The files Number of Umrah permits by user type.xlsx and Umrah permits for the year 1446 by month and type.xlsx were used as the main sources for Umrah permit counts for the Hijri years 1445 AH and 1446 AH [9], [10]. These datasets were monthly, numerical, and temporal, as they included the number of permits by Hijri month and user type. They were selected because they represent the main basis for estimating the number of Umrah performers during the study period.

The files Number of Umrah permits in Ramadan.xlsx and Umrah permits by day for the month of Ramadan 1446.xlsx were also used to represent daily

Umrah data during the month of Ramadan [11], [12]. These files were selected because Ramadan is a high-crowding season, and the availability of daily data for this month helps represent daily changes more accurately compared with relying only on monthly averages.

In addition, Hajj-related data for 1445 AH and 1446 AH was used to represent the possible effect of the Hajj season on crowd levels at Al-Masjid Al-Haram, especially during Dhul-Qi'dah and Dhul-Hijjah. The Hajj-related datasets used in this project included Number of pilgrims from abroad by gender per day.xlsx, Number of domestic pilgrims in 1445 by age.xlsx, Number of pilgrims entering by age group for 1446.xlsx, and Number of domestic pilgrims by age group for 1446.xlsx [13]–[16]. These datasets were seasonal and numerical, as they described pilgrim numbers according to gender, age group, and entry-related information. They were selected because Hajj activities may increase crowd levels around Al-Masjid Al-Haram during specific periods of the Hijri year. These files also helped create Hajj-related features such as Hajj and Tawaf_Ifadah, which were later integrated into the final forecasting dataset.

The dataset Average Maximum Temperature, by Month and Meteorological Station was used as an environmental data source in this project [17]. The dataset was obtained from the Saudi Open Data Portal. It provides monthly average maximum temperature values for different meteorological stations across Saudi Arabia. Since the study area is Al-Masjid Al-Haram in Makkah, the Makkah meteorological station was selected because it is the closest available weather station to the study area. The temperature values from this station were used to create the AvgTemp_C column. In addition, the temperature values were classified into weather categories such as Hot, Warm, Pleasant, and Cold to support the forecasting and recommendation system.

Finally, a Hijri/Gregorian calendar for 1445 AH and 1446 AH was created during the preprocessing stage to link datasets that depend on Hijri dates with datasets that depend on Gregorian dates. This calendar was necessary to integrate all sources into one final dataset containing one daily record for each date.

Overall, these sources helped build the final dataset, **Intelligent System for Predicting Visitor Crowd Levels and Optimal Visiting Times at Al-Masjid Al-Haram.xlsx**, which was later used for analysis and modeling. This file contains the date, Hijri month, weekday, temperature, weather category, number of Umrah performers, Hajj-related features, Tawaf Al-Ifadah feature, and crowd level.

3.3 Raw Data Description

Before starting the preprocessing stage, the data used in this project was distributed across several different Excel files, all of which were related to factors that may affect the number of Umrah performers and crowd levels at Al-Masjid Al-Haram. The nature of these files varied in terms of data type and level of detail. Some files contained monthly data, some included daily data, while others included seasonal Hajj-related data or climate-related data associated with temperature.

In general, the data used in the project was divided into three main types. The first type was numerical data, such as the number of Umrah permits, the number of pilgrims, and average temperature. The second type was temporal data, such as Hijri date, Gregorian date, month, and weekday. The third type was categorical data, such as user type, age group, month name, and weather category after preprocessing.

This diversity in the data was important because it helped build a broader representation of the factors affecting crowding. Umrah permit data represented

the main basis for estimating the number of Umrah performers, Ramadan data helped represent a high-crowding period on a daily basis, while Hajj and weather data added seasonal and environmental factors that may influence visitor movement and crowd levels.

Table 3.1: Description of Raw Datasets Before Preprocessing

File Name	Approx. Records Before Preprocessing	Main Columns	Data Type	Key Columns
Number of Umrah permits by user type.xlsx	60 records	4 columns	Monthly, numerical, categorical	Year, month, user type, number of Umrah permits
Umrah permits for the year 1446 by month and type.xlsx	55 records	4 columns	Monthly, numerical, categorical	Hijri year, month, user type, number of permits
Number of Umrah permits in Ramadan.xlsx	30 records	4 columns	Daily, temporal, numerical	Year, Gregorian day, Hijri day, number of Umrah permits
Umrah permits by day for the month of Ramadan 1446.xlsx	29 records	4 columns	Daily, temporal, numerical	Gregorian year, Gregorian day, Hijri day, number of permits
Number of pilgrims from abroad by gender per day.xlsx	53 records	4 main columns	Daily, numerical, categorical	Year, day, number of females, number of males
Number of domestic pilgrims in 1445 by age.xlsx	6 records	3 main columns	Numerical, categorical	Year, age group, number of domestic pilgrims
Number of pilgrims entering by age group for 1446.xlsx	212 records	3 columns	Temporal, numerical, categorical	Entry date, age group, number of pilgrims
Number of domestic pilgrims by age group for 1446.xlsx	6 records	3 columns	Numerical, categorical	Hijri year, age group, number of pilgrims
Average Maximum Temperature, By Month and Meteorological Station	Monthly data by station	Monthly columns	Climate, numerical	Station name, monthly temperature values

The two Umrah permit files contained monthly data; therefore, they were not directly suitable for daily forecasting before preprocessing. In contrast, the Ramadan datasets contained daily records, which made them more detailed and allowed them to be used directly for Ramadan-related analysis. The Hajj datasets had different structures depending on the data source. Some files contained daily records, such as the data of pilgrims arriving from abroad, while others were aggregated by age group, such as the domestic pilgrim datasets.

In addition, the weather dataset contained monthly average maximum temperature values by meteorological station. For this study, the Makkah meteorological station was selected because it represents the closest available weather station to Al-Masjid Al-Haram and the study area. The temperature values were used to create the AvgTemp_C column and to classify the weather into categories such as Hot, Warm, Pleasant, and Cold.

After preprocessing and integration, a final daily dataset was created under the file name Intelligent System for Predicting Visitor Crowd Levels and Optimal Visiting Times at Al-Masjid Al-Haram.xlsx. This file contains 708 daily records and 12 main columns. Each record represents one day from the study period, while the columns represent the variables used for analysis and modeling.

Table 3.2: Description of Final Dataset Columns After Preprocessing

Column Name	Data Type	Description
Hijri_Year	Numerical / Temporal	Hijri year
Hijri_Date	Temporal	Hijri date
Hijri_Month	Categorical / Temporal	Hijri month name
Gregorian_Date	Temporal	Gregorian date
Weekday_AR	Categorical	Weekday name in Arabic
AvgTemp_C	Numerical	Average temperature
Weather_Category	Categorical	Weather condition category
المعتمرين	Numerical	Daily number of Umrah performers
Hajj	Numerical	Feature representing the effect of the Hajj season
Tawaf_Ifadah	Numerical	Feature representing the effect of Tawaf Al-Ifadah
Crowding_Level	Categorical	Categorical crowding level
Crowding_Level_Num	Numerical	Numerical crowding level

Therefore, before preprocessing, the data was not unified in terms of time frequency and formatting. Some files were monthly, others were daily, and some were seasonal. After preprocessing, the data was transformed into one unified daily dataset suitable for building prediction models for Umrah counts and crowd levels.

3.4 Data Structure and Challenges

The datasets used in this project had heterogeneous structures and different temporal granularities, which created several preprocessing and integration challenges before the forecasting stage. Some datasets were available at the monthly level, while others contained daily observations. In addition, some files depended on Hijri dates, whereas others used Gregorian dates.

One of the main challenges was the difference between monthly and daily data structures. The Umrah permit datasets for 1445H and 1446H were mainly available as monthly totals, while Ramadan datasets contained daily observations. Since the forecasting objective of the project required daily

prediction, the monthly datasets had to be transformed into continuous daily time-series records.

Another challenge was the coexistence of Hijri and Gregorian calendar systems across the datasets. Some files used Hijri months and Hijri dates, while others used Gregorian-based daily records. Therefore, a unified Hijri/Gregorian calendar was created during preprocessing to ensure chronological consistency and correct temporal alignment across all datasets.

The project also faced challenges related to inconsistent column names, formatting differences, and variations in the written representation of Hijri month names. These inconsistencies required standardization before the datasets could be integrated into one unified structure.

In addition, the datasets varied in terms of feature types and aggregation levels. Some files contained numerical temporal records, while others included categorical demographic information or seasonal Hajj-related statistics. Therefore, preprocessing and transformation stages were necessary to unify the data structure and prepare the dataset for forecasting and crowd-level analysis.

Overall, handling these structural and temporal challenges was necessary to create a reliable daily forecasting dataset suitable for machine learning modeling and seasonal crowd analysis.

3.5 Data Integration

Since the project used several datasets from the same official source, it was necessary to integrate them into one unified dataset. Although the files came from the Saudi Open Data Platform, they were not ready to be used together directly because each file had a different structure, time format, and level of detail.

The main integration key used in this project was the date. A Hijri/Gregorian calendar was created to link Hijri dates with Gregorian dates for the years 1445 AH and 1446 AH. This step was essential because some datasets

were based on Hijri dates, while others used Gregorian dates. The calendar helped create a unified daily timeline that allowed all files to be merged correctly.

The Umrah permit datasets were integrated based on the Hijri year and Hijri month, because these files were originally monthly. The Ramadan datasets were integrated using the daily date, because they already contained daily records. The Hajj datasets were linked to the relevant Hijri months, mainly Dhul-Qi'dah and Dhul-Hijjah, since these months are strongly related to Hajj activities. Weather data was integrated based on the month, where the temperature values for Makkah were assigned to the corresponding Hijri months.

During the integration process, several issues were addressed. The first issue was the difference between monthly and daily data, which required converting monthly Umrah data into daily estimates. The second issue was the difference between Hijri and Gregorian date formats, which required calendar mapping. Another issue was the variation in column names and month names across files, which required standardization before merging.

After completing the integration process, all datasets were combined into one final daily dataset. Each row represented one day, and each day included date information, Umrah count, weather information, Hajj-related features, Tawaf Al-Ifadah feature, and crowding level. This integrated dataset was then used as the main input for analysis and prediction models.

3.6 Data Cleaning

Before integrating the datasets and using them for analysis and modeling, several data cleaning steps were performed to improve data quality, standardize the data structure, and reduce issues that could affect the reliability of the project results. Since the data was collected from several different files, the cleaning process was necessary to ensure that all files could be combined into one unified daily dataset.

First, the column names in all files were cleaned by removing extra spaces, unnecessary symbols, and inconsistent formatting. This step helped facilitate data processing, integration, and comparison across the different files. The formats of the Hijri month names were also standardized, as some files used different written forms of Hijri month names, while other files used numerical month values. Therefore, all months were converted into one consistent Hijri month format using custom preprocessing functions.

Text values were also cleaned to reduce minor language-related inconsistencies, such as unnecessary symbols, spacing differences, and variations in Arabic letter forms. This helped reduce matching problems across files, especially when linking records based on Hijri month names, user type, or date-related fields.

In addition, numerical columns were converted into a unified numeric format. Values that could not be converted into numbers were treated as missing or invalid values. Rows containing missing values in essential fields, such as dates or core numerical values, were reviewed to determine whether they should be removed, corrected, or handled during the preprocessing stage.

The cleaning process also included correcting date formats and converting them into a unified structure. Some files used Gregorian dates, while others used Hijri dates. Therefore, a unified Hijri/Gregorian calendar was created and used to link all dates consistently. This step was necessary because the final dataset needed to represent each record as a daily observation containing both Hijri and Gregorian date information.

Regarding duplicate records, the calendar data and daily datasets were checked for repeated records. When duplicates were found, only one record for each date was kept to prevent duplicated values during the integration process. Some invalid or incomplete values, such as empty symbols or text values

appearing inside numerical fields, were also cleaned or excluded before inserting the data into the final dataset.

During the data cleaning and preprocessing stage, some challenges were encountered due to the incomplete availability of direct daily Umrah permit records for Dhul-Qi'dah 1446H. To preserve the continuity of the daily time-series data, estimated daily values were generated using weighted distribution techniques based on historical patterns from Dhul-Qi'dah 1445H. In this process, records related to Saudi citizens and Gulf nationals were selected. These categories were selected because they represent more continuous visitor activity during the pre-Hajj period compared with some international Umrah categories that may be affected by seasonal or regulatory restrictions associated with the Hajj season. After generating these records, they were integrated into the final dataset to preserve temporal completeness and support the stability of the forecasting process.

After completing all cleaning steps, the data became more consistent and organized. This helped create a unified daily dataset and prepared it properly for the transformation stage and the development of prediction models. This stage also helped prepare the data for feature engineering and the development of forecasting models, which will be explained in the following chapters.

3.7 Data Transformation

After completing the data cleaning stage, several data transformation steps were performed to convert the data into a suitable format for analysis and prediction modeling. The main purpose of this stage was to standardize the data structure and prepare it for daily time-series analysis and other analytical processes.

First, the dates were converted into a unified format that could be used for data linking and temporal analysis. Since some files used Hijri dates while others used Gregorian dates, a unified Hijri/Gregorian calendar was created to connect both date systems. New columns were also created to represent the Hijri year, Hijri

month, and Hijri day, which made it easier to handle temporal information throughout the project.

Some textual and numerical columns were also converted into suitable data types. For example, numerical columns such as the number of Umrah performers, number of pilgrims, and temperature values were converted into numeric formats. Meanwhile, date columns were converted into time-based formats that could be processed and analyzed programmatically.

In addition, some columns were renamed and standardized across different files to make the integration and processing stages easier. Hijri month names were also unified into a consistent format to ensure compatibility across all datasets.

The transformation stage also included creating new derived features from the original data. For example, the `Weather_Category` feature was created based on the average temperature, where weather conditions were classified into categories such as Hot, Warm, Pleasant, and Cold. Additional Hajj-related features, such as `Hajj` and `Tawaf_Ifadah`, were also created to represent the effect of the Hajj season and Tawaf Al-Ifadah on crowd levels.

Crowding levels were also transformed into both categorical and numerical forms. The `Crowding_Level` column was created to classify crowding into levels such as Low, Medium, High, and Critical. In addition, the numerical column `Crowding_Level_Num` was created so that crowding levels could be used later in analysis and modeling.

Overall, the data transformation stage helped prepare the final dataset in a unified and organized structure, making it more suitable for time-series analysis and for developing prediction models for crowd levels at Al-Masjid Al-Haram.

3.8 Time Series Preparation

Since this project focuses on predicting crowd levels at Al-Masjid Al-Haram, it was necessary to prepare the dataset as a daily time series. This step was important because forecasting models require chronologically ordered data in order to learn patterns related to days, months, seasons, Ramadan, and Hajj.

First, all records were arranged according to date. The `Gregorian_Date` column was used as the main chronological reference because it provides a clear daily sequence. At the same time, the `Hijri_Date` and `Hijri_Month` columns were

retained because they are important for representing the Hijri calendar and religious seasons such as Ramadan and Dhul-Hijjah.

The data frequency was set to daily, meaning that each row in the final dataset represents one day. This was necessary because the objective of the project is to predict daily Umrah counts and daily crowd levels. Therefore, monthly Umrah data was converted into daily values, while Ramadan daily data was used directly because it was already available at the daily level.

Time gaps were also handled carefully. Some of the original files did not contain complete daily records, especially the monthly Umrah permit files. Therefore, estimated daily values were generated to fill these gaps and create a continuous daily sequence without missing dates. The Hijri/Gregorian calendar also helped ensure that all days in the study period were included and correctly matched between the Hijri and Gregorian date systems.

In addition to the general handling of time gaps, Dhul-Qi'dah 1446H required a specific treatment during the time-series preparation stage due to the incomplete availability of direct daily observations for this month. To avoid discontinuity in the daily time-series structure, estimated records were generated using Dhul-Qi'dah 1445H data as a historical reference. The estimation focused on Saudi citizens and Gulf nationals, as these groups are more likely to reflect stable and continuous movement during the pre-Hajj period compared with some international Umrah categories that may be affected by seasonal or regulatory restrictions. These estimated records were included to preserve the continuity of the daily time-series data and support the stability of the forecasting process.

The data was split chronologically instead of using a random split. Data from 1445H was used for training, while data from 1446H was used for testing. This approach was chosen because it is more suitable for time-series projects,

where the model should be trained on historical data and then tested on a later time period to better represent realistic forecasting.

The المعتمرين / Umrah_Count column was selected as the main target variable for numerical prediction because it represents the daily number of Umrah performers and reflects the volume of movement inside Al-Masjid Al-Haram. In addition, the Crowding_Level_Num column was prepared as another target variable for analyzing or predicting crowd levels, where crowding levels were converted into numerical values as follows: Low = 1, Medium = 2, High = 3, and Critical = 4.

Overall, this stage helped transform the data into an organized daily time series, where each row represents one day and includes temporal, seasonal, environmental, Umrah-related, Ramadan-related, and Hajj-related features. As a result, the dataset became suitable for training forecasting models and analyzing crowd patterns over time.

3.9 Synthetic Data Generation

The synthetic daily data generation stage was one of the essential steps in this project because part of the original data obtained from the Saudi Open Data Platform was not available at a direct daily level. Instead, several Umrah permit datasets were provided as monthly totals. Since the objective of the project is to predict daily Umrah counts and crowd levels at Al-Masjid Al-Haram, it was necessary to convert the monthly data into an organized daily time-series format suitable for forecasting models.

3.9.1 Daily Distribution

First, the monthly Umrah permit data for 1445H and 1446H was used. The total number of permits for each Hijri month was extracted, and an initial daily value was calculated by dividing the monthly total by the number of days in that

Hijri month. This step helped create a continuous daily sequence instead of relying only on monthly records.

3.9.2 Weighted Distribution

However, equal distribution alone was not sufficient because crowding at Al-Masjid Al-Haram is not expected to be the same across all weekdays. Therefore, a weighted distribution system was applied when generating the daily values. Higher weights were assigned to Thursday, Friday, and Saturday compared with the remaining weekdays in order to represent the expected increase in visitor density during weekends. Friday received the highest weight, followed by Thursday and Saturday, while the remaining weekdays kept the basic weight.

Weather was also included as an additional supporting factor in the estimation process. The average temperature was linked to each Hijri month, and the weather category was used as a supporting variable when adjusting daily values. Temperature values were classified into categories such as Cold, Pleasant, Warm, and Hot, which helped include a simplified environmental effect in the generated daily Umrah counts.

3.9.3 Ramadan Handling

Ramadan was handled differently from the other months. When daily Ramadan data was available, it was used directly instead of being generated from monthly totals. This was important because Ramadan is one of the busiest periods at Al-Masjid Al-Haram, and the availability of daily data for this month allowed the project to represent daily variations more accurately than using estimated distribution.

After generating daily values for non-Ramadan months, an important correction step was applied to ensure that the sum of the generated daily values within each month matched the original monthly permit total. This means that

although daily values were adjusted using weights, the final monthly total remained consistent with the original data. This step helped preserve consistency between the source data and the generated daily dataset.

3.9.4 Dhul-Qi'dah 1446H Estimation

A specific synthetic data generation step was also applied for Dhul-Qi'dah 1446H due to the limited availability of complete direct daily observations for this month. Instead of excluding this period or leaving it as a gap in the dataset, Dhul-Qi'dah 1445H data was used as a historical reference to generate estimated daily values. This procedure was applied to preserve the continuity of the time-series structure and ensure the availability of a complete daily sequence that could be used in the analysis and forecasting stages.

In this process, only the permit records related to Saudi citizens and Gulf nationals were selected. These categories were selected because they represent more continuous visitor activity during the pre-Hajj period compared with some international Umrah categories that may be affected by seasonal or regulatory restrictions associated with the Hajj season. Therefore, these categories provided a more reasonable reference for estimating Dhul-Qi'dah 1446H values.

After extracting and aggregating the selected categories from Dhul-Qi'dah 1445H data, the total estimated value was redistributed across the days of Dhul-Qi'dah 1446H using the same weighted daily distribution method. This helped avoid simple equal distribution and allowed the generated values to better reflect expected daily variation, especially the possible increase in visitor movement during weekends. The simulated records were then merged into the final daily dataset to preserve the continuity of the daily time-series data and support the stability of the forecasting process.

3.9.5 Hajj Feature Construction

In addition, estimated Hajj-related features were created, including Hajj and Tawaf_Ifadah. Hajj data was used to represent seasonal effects during Dhul-Qi'dah and Dhul-Hijjah. The Tawaf_Ifadah was generated by distributing the effect of pilgrims over the days from 10 Dhul-Hijjah until the end of the month using decreasing weights. Higher values were assigned to the first days, and then the values gradually decreased in the following days.

3.9.6 Assumptions

This stage was based on several clear assumptions. First, it was assumed that monthly data could be transformed into estimated daily values. Second, it was assumed that weekends usually experience higher crowd density than regular weekdays. Third, it was assumed that the effect of Tawaf Al-Ifadah is stronger in the early days after 10 Dhul-Hijjah and then gradually decreases. Fourth, it was assumed that Saudi citizens and Gulf nationals provide a more stable reference for estimating Dhul-Qi'dah 1446H data during the pre-Hajj period. It was also assumed that monthly average temperature could represent the general weather effect for that month.

3.10 Feature Engineering

After completing data cleaning and transforming the data into a suitable daily format, the feature engineering stage was performed to create new variables that help the model understand the factors affecting the number of Umrah performers and crowd levels at Al-Masjid Al-Haram. Some of these variables were not directly available in the original files; therefore, they were derived from the available data, such as dates, Ramadan data, Hajj data, and

weather data. These engineered features were designed to improve the model's ability to capture temporal patterns, seasonal crowd behavior, and event-driven fluctuations.

The first important factor considered was the effect of Ramadan. Since Ramadan is one of the most crowded periods at Al-Masjid Al-Haram, it was handled separately. When daily Ramadan data was available, it was used directly instead of estimating it from monthly data. This helped represent the daily variation in the number of Umrah performers during Ramadan more accurately compared with other months, which often depended on daily distribution from monthly data.

The Hajj was also created to represent the effect of the Hajj season on crowd levels. This feature was derived from the pilgrim datasets, including both pilgrims coming from abroad and domestic pilgrims. It focused mainly on Dhul-Qi'dah and Dhul-Hijjah, as these months are most strongly associated with the Hajj season. This feature helped the model distinguish between regular days and days that may be affected by large numbers of pilgrims.

In addition, the Tawaf_Ifadah was created to represent the expected impact of Tawaf Al-Ifadah during the Hajj season. This feature was calculated by distributing the total number of pilgrims over the days from 10 Dhul-Hijjah until the end of the month using decreasing weights. Higher weights were assigned to the first days because the impact of Tawaf Al-Ifadah is usually greater after the 10th of Dhul-Hijjah, and then the values gradually decrease in the following days. This feature provided a more specific representation of an important period within the Hajj season.

The weekend effect was also represented by assigning higher weights to Thursday, Friday, and Saturday when generating daily values from monthly data. This assumption was based on the actual nature of visitor movement, as weekends

usually experience higher Umrah density compared with regular weekdays. Therefore, these weights helped make the daily distribution of Umrah counts more logical compared with a simple equal distribution.

Regarding weather, the `Weather_Category` feature was created based on the average temperature in Makkah. Temperature values were classified into categories such as Cold, Pleasant, Warm, and Hot. This classification transformed weather data from a numerical value into a categorical feature that is easier to interpret and can help the model understand the potential effect of climatic conditions on visitor movement.

Finally, crowding-related variables were created from the daily Umrah count column. The daily number of Umrah performers was classified into four categories: Low, Medium, High, and Critical, and stored in the `Crowding_Level` column. These categories were then converted into numerical values in the `Crowding_Level_Num` column, using the following encoding: Low = 1, Medium = 2, High = 3, and Critical = 4. This transformation was necessary because machine learning models require numerical representations of crowd levels.

Overall, the feature engineering stage helped make the dataset more capable of representing the seasonal, temporal, and environmental conditions related to Al-Masjid Al-Haram. These features provided the models with additional information, which helped improve the representation of temporal and seasonal patterns within the project dataset.

3.11 Crowding Level Construction

The crowding level classification was created to transform the numerical Umrah count into interpretable crowd-density categories that can support forecasting analysis, dashboard visualization, and recommendation generation.

The daily number of Umrah performers was divided into four crowding categories: Low, Medium, High, and Critical. These categories were designed to provide a simplified representation of expected crowd intensity at Al-Masjid Al-Haram. The classification process helped convert continuous numerical

forecasting outputs into operational crowd-level indicators that are easier to interpret within the recommendation dashboard.

After defining the categorical crowding levels, numerical encoding was applied to make the values suitable for machine learning analysis and modeling. The encoding process was performed as follows: Low = 1, Medium = 2, High = 3, and Critical = 4.

The resulting numerical feature, Crowding_Level_Num, was later used in forecasting analysis and visualization processes. This transformation also helped connect numerical forecasting outputs with recommendation indicators such as expected crowd intensity and visiting suitability.

Overall, the crowding level construction stage improved the interpretability of the forecasting system and helped bridge the gap between numerical prediction outputs and practical crowd-analysis insights

Table 3.3: Final Dataset Columns

Column Name	Data Type	Description
Hijri_Year	Numerical / Temporal	Hijri year
Hijri_Date	Temporal	Hijri date
Hijri_Month	Categorical / Temporal	Hijri month name
Gregorian_Date	Temporal	Gregorian date
Weekday_AR	Categorical	Weekday name in Arabic
AvgTemp_C	Numerical	Average temperature
Weather_Category	Categorical	Weather condition category
المعتمرين	Numerical	Daily number of Umrah performers
Hajj	Numerical	Feature representing the effect of the Hajj season
Tawaf_Ifadah	Numerical	Feature representing the effect of Tawaf Al-Ifadah
Crowding_Level	Categorical	Categorical crowding level
Crowding_Level_Num	Numerical	Numerical crowding level

Table 3.4: Example of the Final Dataset Structure

Field	Record 1	Record 2	Record 3
Hijri_Year	1445	1445	1445
Hijri_Date	01-01-1445	02-01-1445	03-01-1445
Hijri_Month	محرم	محرم	محرم
Gregorian_Date	2023-07-19	2023-07-20	2023-07-21
Weekday_AR	الأربعاء	الخميس	الجمعة
AvgTemp_C	43.3	43.3	43.3

Weather_Category	Hot	Hot	Hot
المعتمرين	38390	47987	51825
Hajj	0	0	0
Tawaf_Ifadah	0	0	0
Crowding_Level	Medium	Medium	Medium
Crowding_Level_Num	2	2	2

This final dataset was used to build the prediction models. The المعتمرين / Umrah_Count column was used as the target variable for numerical prediction of daily Umrah performers, while Crowding_Level_Num was used when predicting or analyzing numerical crowd levels. Therefore, the final dataset became suitable for analyzing temporal and seasonal patterns and training the prediction models developed in this project.

3.12 Final Dataset

After completing the stages of data collection, cleaning, transformation, integration, and feature engineering, the final dataset used for analysis and prediction modeling was produced. The final file was saved as: Intelligent System for Predicting Visitor Crowd Levels and Optimal Visiting Times at Al-Masjid Al-Haram.xlsx.

This file represents the final version prepared for modeling. It contains daily data covering the Hijri years 1445 AH and 1446 AH. After preprocessing, the dataset contained 708 daily records and 12 columns. Each row represents one day, while the columns represent temporal, environmental, seasonal, and crowd-related features.

The final dataset was structured as a daily time series. Each daily record includes the Hijri and Gregorian dates, Hijri month, weekday, average temperature, weather category, daily Umrah count, Hajj-related features, Tawaf Al-Ifadah feature, and both categorical and numerical crowding levels.

This final dataset was used to build the prediction models. The المعتمرين / Umrah_Count column was used as the target variable for numerical prediction of daily Umrah performers, while Crowding_Level_Num was used when predicting or analyzing numerical crowd levels. Therefore, the final dataset became suitable for analyzing temporal and seasonal patterns and training the prediction models developed in this project.

3.13 Limitations of the Data

Although this project relied on official data from reliable sources, the dataset used in the study had several limitations that affected the nature of the analysis and modeling. Documenting these limitations is important because they represent an essential part of the academic evaluation of any data-driven project.

One of the main limitations of this project is that part of the daily data used was not directly observed data. Instead, it was generated using estimation methods based on the available monthly data. This was due to the lack of complete daily records for the number of Umrah performers across all months in the study period. Therefore, estimation methods and weighted daily distribution techniques were used to convert monthly data into a daily time series suitable for modeling. Although these methods helped create a continuous daily dataset, they remain approximate estimates and may not fully reflect real daily variations.

Another specific limitation is related to Dhul-Qi'dah 1446H, where complete direct daily observations were not fully available for this month. Therefore, estimated daily values were generated using Dhul-Qi'dah 1445H data as a historical reference, with a focus on Saudi citizens and Gulf nationals. These categories were selected because they represent more continuous visitor activity during the pre-Hajj period compared with some international Umrah categories that may be affected by seasonal or regulatory restrictions associated with the Hajj season. Although this approach helped preserve the continuity of the daily

time-series dataset, the generated values should be interpreted as estimated records rather than direct observed daily data.

The project also relied on several assumptions during the generation of daily data, such as assigning higher weights to certain days, including Thursday, Friday, and Saturday, based on the assumption that these days usually experience higher visitor density compared with regular weekdays. Although these assumptions are based on the expected nature of visitor movement, they may not accurately represent actual behavior during all periods.

Another limitation is that some variables, such as weather data, were available only at the monthly level rather than the daily level. This means that daily weather variations were not fully represented in the dataset.

There is also a possibility of bias resulting from the method used to generate daily data or from relying only on the available data. For example, some patterns in the dataset may become smoother than reality due to the use of distribution or interpolation methods, which could affect the performance of prediction models or make some results less representative of real daily changes.

Overall, despite these limitations, the data was handled with as much organization and documentation as possible. All assumptions and preprocessing steps were clearly explained to ensure academic transparency and improve the reliability of the project results.

3.14 Chapter Summary

This chapter presented the data collection and preparation process used to build the dataset for the proposed Umrah crowd forecasting system. The chapter described the main data sources, including Umrah permit data, Ramadan data, Hajj-related data, weather data, and the Hijri/Gregorian calendar used to align dates across all sources.

The chapter also explained the preprocessing stages required to convert heterogeneous raw files into a unified daily dataset. These stages included data

cleaning, transformation, integration, time-series preparation, synthetic daily data generation, feature engineering, crowding level construction, and final dataset preparation.

A special treatment was applied to Dhul-Qi'dah 1446H because complete direct daily records were not fully available for this month. Estimated values were generated using historical data from Dhul-Qi'dah 1445H, focusing on Saudi citizens and Gulf nationals to preserve the continuity of the daily time series.

By the end of this chapter, the final dataset was prepared as a daily time series containing 708 records and 12 main columns. This dataset became suitable for forecasting Umrah performer counts, analyzing crowd levels, and supporting the dashboard-based recommendation system discussed in the following chapters.

Chapter 4: Methodology and Implementation

4.1 Overview of the Methodology and Implementation

This chapter presents the methodology and system design followed to develop the intelligent forecasting system for predicting Umrah performer crowd levels and supporting suitable Umrah-time selection at Al-Masjid Al-Haram. The methodology was designed as an end-to-end workflow that starts from the prepared daily Umrah dataset, applies time-series feature engineering, trains and evaluates the forecasting model, and finally presents the outputs through an interactive dashboard supported by a recommendation component.

The system was developed to support data-driven crowd analysis rather than only producing numerical predictions. Therefore, the methodology combines four main components: prepared data utilization, feature engineering, machine-learning forecasting, and dashboard-based recommendation support. The forecasting component focuses on predicting the expected number of Umrah performers using a final adopted machine-learning model, while the dashboard component transforms these outputs into understandable crowd-level indicators and visiting-day recommendations.

Since the project deals with time-dependent Umrah performer patterns, the data was split chronologically rather than randomly. Data from Hijri year 1445H was used for training, while data from Hijri year 1446H was used for testing. This approach is suitable for time-series forecasting because it reflects a realistic forecasting scenario, where the model learns from historical data and is evaluated on a later period.

After conducting forecasting experiments and model comparisons, the final adopted forecasting approach was the **Standalone Growth-Rate XGBoost model**. Instead of directly predicting the number of Umrah performers, this model predicts the weekly growth rate of Umrah performers using the `growth_rate_7` target. The predicted growth rate is then reconstructed into the expected number of Umrah performers using the previous weekly value, `lag_7`.

The target variable used for training the final model is calculated as:

$$\text{growth_rate_7} = (\text{Umrah_Count} - \text{lag_7}) / \text{lag_7}$$

After prediction, the final expected number of Umrah performers is reconstructed as:

$$\text{Predicted_Umrah_Count} = \text{lag_7} \times (1 + \text{Predicted_Growth_Rate_7})$$

This approach allows the model to learn the percentage change in Umrah performer numbers compared with the same weekly pattern, instead of only learning the raw visitor count. This is useful because Umrah movement is affected by weekly behavior, Ramadan, Hijri months, and seasonal changes.

The final outputs are presented through an interactive dashboard. The dashboard displays the selected day, predicted Umrah performer count, expected crowding level, temperature, final recommendation, and Hajj-related indicators when applicable. It also provides a seven-day forecasting plot and allows the user to optionally display the best available day within the nearby seven-day period.

Overall, the methodology follows a complete pipeline consisting of prepared daily data input, feature engineering, growth-rate forecasting, predicted count reconstruction, crowd-level classification, dashboard visualization, and recommendation-based decision support.

Table 4.1: Main Components of the Proposed Methodology

Component	Purpose	Main Output
Prepared data utilization	Uses the final daily Umrah dataset as the input for the forecasting workflow.	Organized daily time-series data.
Feature utilization	Applies temporal, seasonal, weather-related, and religious-event features.	Model-ready feature set.
Forecasting model	Predicts Umrah performer counts using the Standalone Growth-Rate XGBoost model.	Predicted Umrah performer count.
Performance evaluation	Evaluates forecasting accuracy using MAE, RMSE, and MAPE.	Numerical model performance results.
Dashboard and recommendation	Transforms model outputs into visual indicators and visiting-day guidance.	Crowd level, suitability decision, and best day within seven days when needed.

4.2 Overall System Architecture

The overall system architecture describes how the proposed system moves from prepared daily data to practical recommendation outputs. The system is designed as a connected pipeline that begins with the prepared daily Umrah dataset and ends with a dashboard that helps Umrah performers understand expected crowding conditions and choose a more suitable day when needed.

The first stage of the architecture is the prepared daily dataset. This dataset contains Hijri and Gregorian dates, weekday information, average temperature, daily Umrah performer counts, Hajj-related indicators, Tawaf Ifadah-related indicators, and crowding level information. The dataset was prepared in Chapter 3 and used as the foundation for model development and dashboard output generation.

The second stage is feature engineering. In this stage, the system creates time-series features such as lag values, rolling means, rolling standard deviations, and difference features. Calendar-based features such as day, month, day of week, and day of year are also generated. In addition, a Ramadan indicator is created to help the model distinguish Ramadan from normal days. Weather-related features are also included when available.

The third stage is the forecasting model. The final adopted model is the Standalone Growth-Rate XGBoost model. The model predicts `growth_rate_7`, which represents the percentage change in Umrah performer counts compared with the value from seven days earlier. The predicted growth rate is then converted into the expected number of Umrah performers using `lag_7`.

The fourth stage is crowd-level classification. After generating the predicted Umrah performer count, the system classifies the expected crowding level into interpretable categories such as Low, Medium, and High. This helps transform numerical predictions into simple indicators that can be understood by non-technical users.

The fifth stage is dashboard visualization. The dashboard presents the predicted Umrah performer count, crowding level, temperature, selected Hijri date, weekday, and Hajj-related indicators when applicable. It also displays a seven-day forecast plot that shows the expected crowd trend for the selected day and the following nearby days.

The final stage is the recommendation component. The system generates a recommendation based on the expected crowding level. If the predicted level is low, the selected day is considered suitable. If the predicted level is medium, the system recommends visiting with caution and choosing a suitable time. If the predicted level is high, the system recommends choosing another day. In addition, the user can optionally press the “best day” button to display the least crowded day within the nearby seven-day forecast window.

Therefore, the system does not only produce numerical forecasts. It converts forecasting outputs into clear indicators and practical recommendations that support Umrah performers in planning their visit to Al-Masjid Al-Haram.

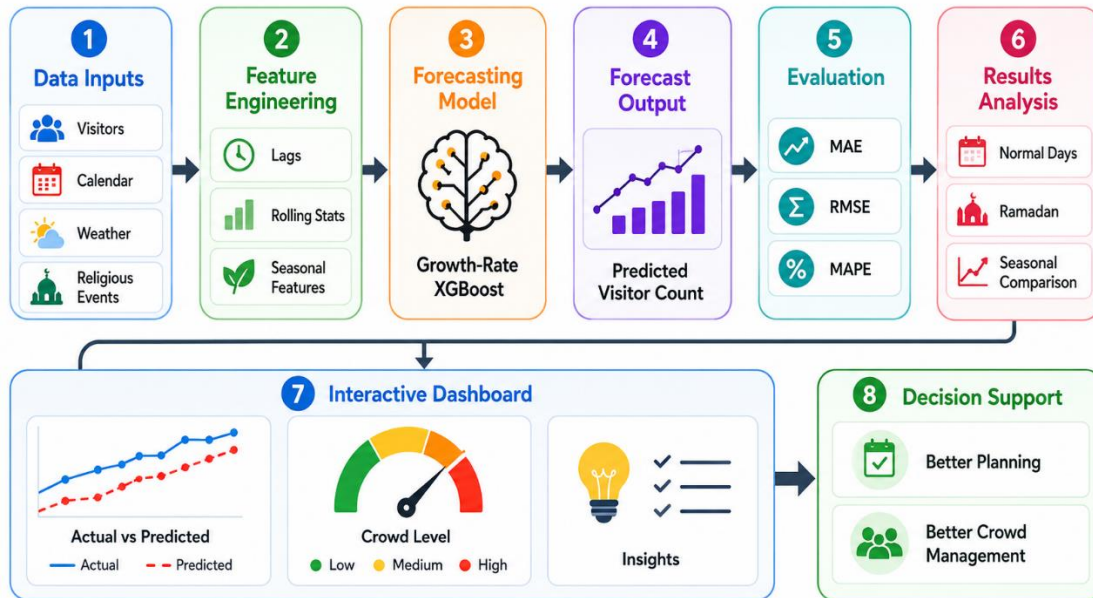


Figure 4.2: Overall Architecture of the Umrah Performer Forecasting and Recommendation System

Layer	Description	Function in the System
Input layer	Prepared daily Umrah dataset with temporal, seasonal, weather, and crowd-related information.	Provides the main data source for forecasting.
Feature layer	Uses lag variables, rolling statistics, calendar features, and religious-season indicators.	Captures temporal and seasonal patterns.
Forecasting layer	Applies the Standalone Growth-Rate XGBoost model.	Predicts expected Umrah performer counts.
Evaluation layer	Uses MAE, RMSE, and MAPE with seasonal analysis.	Measures forecasting accuracy and stability.
Dashboard layer	Displays predictions, crowd levels, and supporting indicators.	Turns results into visual and interpretable outputs.
Recommendation layer	Searches within a nearby seven-day period when crowding is medium or high.	Suggests a less crowded alternative day when needed.

4.3 Data Flow and Processing Pipeline

This section explains how data flows through the proposed forecasting and recommendation system, starting from the prepared dataset and ending with forecasting outputs, dashboard visualization, analytical insights, and

recommendation-based decision support. The processing pipeline was designed to ensure that the data passes through organized stages, including preprocessing, feature engineering, forecasting, evaluation, dashboard integration, and recommendation generation.

The pipeline begins by importing the prepared daily Umrah performer dataset along with the related temporal features. The dataset is arranged chronologically using the `Gregorian_Date` column to ensure that all time-dependent features are generated correctly. The data is then processed through cleaning and transformation stages to ensure consistency, correct formatting, and suitability for forecasting tasks.

After preprocessing, several engineered features are generated to capture temporal patterns, weekly trends, seasonal effects, and growth-rate behavior. These features include lag-based features such as `lag_1`, `lag_7`, `lag_14`, and `lag_30`; rolling statistical features such as `rolling_mean_7`, `rolling_mean_14`, `rolling_mean_30`, `rolling_std_7`, `rolling_std_14`, and `rolling_std_30`; difference features such as `diff_1`, `diff_7`, `diff_14`, and `diff_30`; and calendar-based features such as `day`, `month`, `dayofweek`, and `dayofyear`. The feature `is_ramadan` is also included to help the model distinguish Ramadan from normal days due to the unique crowd behavior during this peak season.

Weather-related features are included when available, such as `AvgTemp_C` and weather-category variables. Meanwhile, `Hajj` and `Tawaf>Ifadah` are preserved as interpretation and recommendation indicators in the dashboard rather than being used as direct training features in the final model.

The processed dataset is then passed to the forecasting layer, where different forecasting models were trained and evaluated during the experimentation phase. These experiments included baseline models, Random Forest models, hybrid forecasting experiments, and XGBoost-based models. Based on the model development process, the **Standalone Growth-Rate XGBoost model** was selected as the final forecasting model.

The final model predicts `growth_rate_7`, which represents the percentage change between the current Umrah performer count and the count from seven days earlier. The target variable is calculated as:

$$\text{growth_rate_7} = (\text{Umrah_Count} - \text{lag_7}) / \text{lag_7}$$

After prediction, the model reconstructs the final daily count using lag_7 as follows:

$$\text{Predicted_Umrah_Count} = \text{lag}_7 \times (1 + \text{Predicted_Growth_Rate}_7)$$

This design allows the system to model relative weekly movement rather than depending only on raw count prediction. This is useful because Umrah crowd movement often follows weekly and seasonal patterns.

Finally, the generated forecasting outputs and analytical insights are integrated into the interactive dashboard and recommendation layer. The dashboard visualizes the selected day, predicted count, expected crowd level, temperature, religious-season indicators when applicable, the seven-day forecast plot, and the optional best-day suggestion. The seven-day forecast window starts from the selected date and continues chronologically for the next six days, even when the window moves from one Hijri month to another.

4.4 Forecasting Model Development

The model development stage focused on building a forecasting model capable of estimating daily Umrah performer counts using historical patterns and time-based features extracted from the dataset. Several models were tested during the experimentation phase, including baseline models, Random Forest models, hybrid forecasting experiments, and XGBoost-based models. After comparing the performance of these models, the **Standalone Growth-Rate XGBoost** model was selected as the final forecasting model because it achieved better performance and was suitable for integration into the interactive dashboard.

The model was developed using the **XGBoost Regression** algorithm. This algorithm is suitable for the project because it can handle nonlinear relationships between the input features and the target variable. This capability is important for forecasting Umrah performer counts because crowd movement is affected by several interacting factors, such as weekly patterns, Ramadan, seasonal changes, previous Umrah performer counts, and environmental factors such as temperature.

The dataset was divided chronologically instead of randomly because the project is based on time-series forecasting. Therefore, data from Hijri year **1445H** was used for training the model, while data from Hijri year **1446H** was used for testing. This split helps simulate a realistic forecasting scenario, where the model learns from past data and is then evaluated on a future period.

The final model was not trained only on the direct Umrah performer count. Instead, it was trained on the **growth_rate_7** target, which represents the rate of change in Umrah performer counts compared with the count from seven days earlier. This target is calculated using the following equation:

$$\text{growth_rate_7} = (\text{Umrah_Count} - \text{lag_7}) / \text{lag_7}$$

After the model predicts **Predicted_Growth_Rate_7**, the expected number of Umrah performers is reconstructed using **lag_7** as follows:

$$\text{Predicted_Umrah_Count} = \text{lag_7} \times (1 + \text{Predicted_Growth_Rate_7})$$

This approach helps the model understand relative changes in Umrah performer numbers instead of relying only on direct raw count prediction. It is also suitable for the nature of Umrah data because Umrah movement often follows weekly and seasonal patterns.

The model development process followed several steps. First, the final daily dataset was loaded and arranged chronologically according to the Gregorian date. Then, time-based features were generated, including lag features, rolling means, rolling standard deviations, and difference features. Calendar features such as day, month, day of week, and day of year were also created, along with the Ramadan indicator. After that, the target variable **growth_rate_7** was constructed. The XGBoost model was then trained using 1445H data and tested using 1446H data. Finally, the trained model and prediction results were saved for use in the dashboard.

The final output of the model is the predicted number of Umrah performers for each day. This output is later used to determine the expected crowding level and display the appropriate recommendation in the dashboard.

Table 4.3: Forecasting Model Development Summary

Stage	Description	Reason for Inclusion
Baseline models	Naive lag-based forecasting approaches such as lag_1 and lag_7.	Provide simple benchmarks for comparison.
Machine learning models	Models trained using temporal, seasonal, and engineered features.	Capture nonlinear relationships and seasonal behavior.
Hybrid experiments	Growth-rate predictions blended with lag-based predictions using alpha.	Test whether combining models improves performance.
Final model	Standalone Growth-Rate XGBoost.	Selected for best balance of accuracy, stability, interpretability, and dashboard suitability.
Evaluation strategy	Chronological split using 1445H for training and 1446H for testing.	Reflects realistic forecasting conditions.

4.5 Feature Engineering and Time-Series Modeling

To improve forecasting performance, the prepared dataset was organized as a chronologically ordered daily time-series dataset. This structure allowed the forecasting model to learn Umrah performer patterns based on previous observations, temporal dependencies, weekly behavior, and seasonal changes. Feature engineering was an important stage in the proposed system because Umrah crowd density is influenced not only by previous performer counts, but also by calendar patterns, religious seasons, and special event-related effects.

Several temporal and seasonal features were generated to help the model capture different aspects of Umrah behavior. These features included lag variables such as lag_1, lag_7, lag_14, and lag_30, which represent Umrah performer counts from previous days. These variables helped the model learn short-term, weekly, biweekly, and monthly dependencies within the time series.

In addition, rolling statistical features were created, including moving averages and moving standard deviations. The rolling mean features helped capture recent crowd trends, while the rolling standard deviation features helped represent fluctuations and instability in Umrah performer numbers over short-term and medium-term periods.

Difference-based features were also included to represent changes in Umrah performer numbers across different time intervals. These features included diff_1, diff_7, diff_14, and diff_30. They helped the model understand day-to-day and week-to-week movement patterns, which are important for detecting increases or decreases in crowd density over time.

Additional calendar-based features were extracted from the date information. These included day, month, dayofweek, and dayofyear. These features helped the forecasting model capture repeated temporal patterns associated with weekdays, monthly changes, seasonal transitions, and religious periods.

The is_ramadan feature was included because Ramadan represents a peak season with unique crowd behavior. This feature allowed the model to distinguish Ramadan days from normal days and helped improve the model's ability to understand seasonal crowd changes.

Weather-related features were also used when available. The numerical temperature feature AvgTemp_C provided environmental context that may support the model's interpretation of crowd behavior. Weather-category variables could also be included as dummy variables when available in the dataset.

The lag_7 feature played a key role in the Growth-Rate XGBoost forecasting approach. It was used as the main reference for calculating the weekly growth-rate target by comparing the current Umrah performer count with the performer count from the previous week. The target variable was calculated as:

$$\text{growth_rate_7} = (\text{Umrah_Count} - \text{lag_7}) / \text{lag_7}$$

After the model predicted the expected growth rate, the lag_7 value was reused to reconstruct the final predicted Umrah performer count:

$$\text{Predicted_Umrah_Count} = \text{lag_7} \times (1 + \text{Predicted_Growth_Rate_7})$$

This allowed the model to focus on relative weekly changes rather than directly predicting raw performer numbers.

The Hajj and Tawaf_Ifadah variables were preserved as interpretation and recommendation indicators in the dashboard. They were used to explain religious-season effects when applicable, especially during Dhul-Qi'dah and Dhul-Hijjah, but they were not used as direct training features in the final adopted model. This separation allowed the model to remain focused on time-series forecasting while still enabling the dashboard to provide useful contextual explanations.

Furthermore, crowd-level features were used to improve interpretability within the dashboard and recommendation layer. The Crowding_Level feature represented the qualitative crowd category, while Crowding_Level_Num provided a numerical representation for analytical processing and visualization. These features helped transform forecasting outputs into understandable crowd indicators that support dashboard-based analysis and Umrah-time recommendations.

Overall, the feature engineering and time-series modeling process enabled the forecasting system to capture historical trends, weekly dependencies, seasonal fluctuations, and religious-event-related crowd effects. This improved the system's ability to model Umrah behavior and supported the final forecasting, dashboard, and recommendation components.

Table 4.4: Main Features Used in Time-Series Modeling

Feature Group	Examples	Purpose
Lag features	lag_1, lag_7, lag_14, lag_30	Represent previous Umrah performer counts.
Rolling statistics	rolling_mean_7, rolling_mean_14, rolling_mean_30, rolling_std_7, rolling_std_14, rolling_std_30	Capture short-term and medium-term trends and fluctuations.
Difference features	diff_1, diff_7, diff_14, diff_30	Represent changes in Umrah performer counts over time.
Calendar features	day, month, dayofweek, dayofyear	Capture repeated temporal patterns.
Seasonal features	is_ramadan	Represent Ramadan-related crowd behavior.
Weather features	AvgTemp_C, weather dummy variables if available	Provide environmental context for crowd behavior.

4.6 Interactive Dashboard and Recommendation Implementation

After producing the forecasting outputs, the final results were integrated into an interactive dashboard designed to support the visual analysis of Umrah performer crowd levels. The dashboard represents the presentation layer of the proposed system, where the forecasting results are transformed into clear and understandable visual information for users.

The dashboard was designed to display the main outputs of the system in a simple and practical way. These outputs include the selected Hijri date, weekday, predicted number of Umrah performers, expected crowding level, temperature, and final recommendation. This allows the user to quickly understand the expected crowd condition for the selected day.

Several dashboard components were developed to support the interpretation of the forecasting results. These components include selected-day information cards, the predicted Umrah performer count, the crowd-level indicator, the temperature indicator, the final recommendation card, and a seven-day forecast plot. The seven-day forecast plot starts from the selected date and continues chronologically for the next six days, even if the forecast window moves from one Hijri month to another.

The dashboard also displays contextual information that supports the recommendation process. For example, the expected temperature is shown because it may affect the comfort of the Umrah performer during the visit. In addition, the system indicates whether the selected day is associated with special

religious-season effects such as Hajj or Tawaf Ifadah when applicable. These indicators help explain possible increases or decreases in expected crowd levels, especially during Dhul-Qi'dah and Dhul-Hijjah.

The dashboard includes a recommendation component that helps the Umrah performer decide whether the selected day is suitable for performing Umrah. The recommendation is based mainly on the predicted number of Umrah performers and the expected crowding level. If the expected crowding level is low, the system indicates that the selected day is suitable. If the expected crowding level is medium, the system recommends visiting with caution and choosing a suitable time. If the expected crowding level is high, the system recommends choosing another day.

In addition, the dashboard provides an optional best-day suggestion. When the user requests a better alternative, the system searches within the nearby seven-day forecast window and suggests the day with the lowest expected number of Umrah performers, excluding the originally selected day. This feature helps the user compare nearby days and choose a less crowded option when needed.

The recommendation system helps transform forecasting results into practical guidance. Instead of only presenting numerical predictions, the dashboard explains the expected crowd condition, the temperature, and any Hajj- or Tawaf Ifadah-related effect. This makes the output more meaningful and easier to use for decision support.

Overall, the dashboard was not designed only to display model outputs, but also to make the forecasting results more understandable and useful for Umrah performers. By combining machine learning forecasting, visual presentation, recommendation logic, temperature indicators, and religious-season indicators, the proposed system provides a practical framework for understanding Umrah crowd patterns and helping Umrah performers choose more suitable days for performing Umrah at Al-Masjid Al-Haram.

Table 4.5: Dashboard Outputs and Recommendation Indicators

Output / Indicator	Description	Use in Recommendation
Predicted Umrah count	Expected number of Umrah performers for the selected day.	Main basis for crowd estimation.
Expected crowding level	Crowd category such as low, medium, high, or critical.	Determines whether the selected day is suitable.
Temperature information	Expected temperature and weather category.	Supports comfort-related interpretation.

Output / Indicator	Description	Use in Recommendation
Seasonal indicators	Hajj and Tawaf Ifadah indicators when applicable.	Explains special religious-season effects.
Best day within seven days	Alternative day with lower expected crowding when needed.	Helps the user choose a more suitable visiting day.

4.7 System Implementation Environment

The proposed forecasting system was implemented using Python as the primary development language because of its strong support for data preprocessing, machine learning, time-series forecasting, visualization, and interactive analytics. The implementation process combined several libraries and analytical tools to support the complete forecasting pipeline, starting from data preparation and feature engineering, and ending with model evaluation, dashboard visualization, and recommendation generation.

Data preprocessing and feature engineering operations were mainly performed using Pandas and NumPy. These libraries were used to handle the daily time-series dataset, organize the data chronologically, generate lag-based features, calculate rolling statistical features, and prepare the forecasting dataset for model training and evaluation.

The machine learning forecasting models were implemented using Scikit-learn and XGBoost. Scikit-learn was used for machine learning experimentation, evaluation workflows, and performance metric calculations, while the XGBoost framework was used to develop the final adopted Standalone Growth-Rate XGBoost forecasting model. This model was selected due to its strong forecasting performance and its ability to capture nonlinear temporal relationships in Umrah patterns.

Visualization and analytical evaluation were supported using Python-based visualization tools such as Matplotlib. These tools were used to generate forecasting plots, model evaluation visuals, seasonal comparisons, and analytical figures that helped interpret model behavior and forecasting performance.

The interactive dashboard layer was developed using Streamlit. Streamlit was selected because it allows Python-based machine learning outputs to be presented through an interactive and user-friendly interface. The dashboard displays the predicted Umrah performer count, expected crowding level, weather-related information, and Hajj and Tawaf Ifadah indicators when applicable. It also presents recommendation outputs that help the Umrah performer understand

whether the selected day is suitable for performing Umrah or whether a better day with lower expected crowding should be considered within a nearby 7-day period.

The implementation and experimentation environment relied mainly on Google Colab and Jupyter-based workflows for model development, experimentation, evaluation, visualization, and result generation. This environment enabled flexible code execution and supported integration between data preprocessing, model training, forecasting evaluation, dashboard development, and recommendation logic.

Overall, the implementation environment provided a flexible and practical framework for developing the complete intelligent forecasting system. It supported all major stages of the project, including preprocessing, feature engineering, forecasting experimentation, model evaluation, interactive visualization, and recommendation-based decision support for Umrah crowd analysis at Al-Masjid Al-Haram.

Table 4.6: Implementation Tools and Their Roles

Tool / Library	Role in the Project	Stage Supported
Python	Primary programming language used to develop the system.	Full project implementation.
Pandas	Data handling, cleaning, transformation, and time-series preparation.	Preprocessing and feature engineering.
NumPy	Numerical processing and array-based operations.	Data preparation and modeling support.
Scikit-learn	Model experimentation, evaluation workflow, and metric calculations.	Model development and evaluation.
XGBoost	Final Standalone Growth-Rate XGBoost forecasting model.	Forecasting model.
Matplotlib	Forecasting plots, comparison visuals, and analytical figures.	Visualization and analysis.
Streamlit	Interactive dashboard and recommendation interface.	Dashboard deployment.
Google Colab / Jupyter	Experimentation, model development, and result generation.	Development environment.

4.8 Chapter Summary

This chapter presented the methodology and system design of the proposed Umrah performer forecasting and recommendation system. It explained the complete workflow starting from the prepared daily dataset, followed by feature

utilization, forecasting model development, model evaluation, dashboard visualization, and recommendation-based decision support.

The chapter also described the overall system architecture and explained how the Standalone Growth-Rate XGBoost model was used as the main forecasting component. The model predicts changes in Umrah performer counts based on temporal and seasonal patterns, while the dashboard transforms these outputs into clear crowding indicators and user-oriented recommendations.

In addition, the chapter explained the role of the recommendation component, which helps users identify whether the selected day is suitable for performing Umrah or whether a better alternative day within a nearby seven-day period should be considered. Overall, the methodology demonstrates how machine learning forecasting, time-series feature engineering, visual analytics, and recommendation logic are combined to support Umrah crowd analysis at Al-Masjid Al-Haram.

Chapter 5: Results and Discussion

5.1 Introduction

This chapter presents the experimental results and discussion of the proposed intelligent forecasting and crowd-analysis system. The chapter begins by evaluating the developed forecasting models using numerical performance metrics, comparative tables, and visual analytical results. Different forecasting approaches, including baseline models, machine learning models, and hybrid forecasting experiments, are compared to identify the most effective approach for predicting Umrah performer counts and crowd levels.

The chapter also includes a seasonal analysis focused on Ramadan due to its highly dynamic crowd behavior and increased Umrah density levels. In addition, the chapter presents dashboard-based analytical results and demonstrates how forecasting outputs can be transformed into interpretable operational insights that support crowd analysis and Umrah-time recommendations.

Finally, the chapter discusses the major findings obtained from the experimentation process by explaining why the selected forecasting model achieved better performance, why Ramadan was more difficult to predict, and what strengths and limitations exist within the proposed system. Overall, this chapter connects the numerical forecasting results with analytical interpretation to evaluate the effectiveness of the proposed system for crowd forecasting and decision support at Al-Masjid Al-Haram.

5.2 Experimental Results

This section presents the forecasting experiments conducted to evaluate the proposed system. The experimental process started with baseline and initial models, followed by hybrid forecasting models, and finally the adoption of the

Standalone Growth-Rate XGBoost model based on its stronger testing performance.

5.2.1 Baseline and Initial Model Results

Baseline forecasting models were initially evaluated to provide benchmark references for comparing the performance of more advanced machine learning forecasting approaches. Several lag-based forecasting strategies were tested using historical Umrah performer patterns from Hijri year 1445H as training data and Hijri year 1446H as testing data.

Model	MAE	RMSE	MAPE
Naive lag_1	6,303.46	13,927.31	12.14%
Naive lag_7	7,484.14	21,556.48	17.46%
Naive lag_14	14,041.29	30,405.19	34.64%
Naive lag_30	28,748.46	45,996.94	80.37%

Table 5.1: Overall baseline forecasting performance comparison.

The baseline results show that Naive lag_1 achieved the lowest error among the baseline approaches. This indicates that the previous day value was a stronger short-term reference than longer lag intervals such as lag_14 and lag_30. However, the baseline models remained limited because they could not fully capture nonlinear seasonal behavior and sudden crowd fluctuations.

5.2.2 Hybrid Model Results

At this stage, hybrid models were tested to examine whether combining machine learning predictions with a simple lag-based prediction could improve

the forecasting accuracy of Umrah performer counts. Two hybrid models were evaluated: Hybrid XGBoost and Hybrid Random Forest.

The hybrid approach combined two prediction components. The first component was the Growth-Rate model prediction, which was designed to capture the weekly change in Umrah performer numbers and used lag_7 to reconstruct the predicted performer count. The second component was the Naive lag_1 prediction, which used the number of performers from the previous day as the forecast for the current day. These two predictions were then combined using a weighting coefficient called alpha.

The hybrid prediction was calculated as follows: Hybrid Prediction = $\alpha \times \text{Growth-Rate Prediction} + (1 - \alpha) \times \text{Naive lag_1 Prediction}$.

To avoid data leakage, the alpha value was not selected using the final test data. Instead, the 1445H dataset was used as development data and was split chronologically into a training subset and a validation subset. The training subset was used to train the model, while the validation subset was used to select the best alpha value. The 1446H dataset was kept unseen and used only for final testing.

Several alpha values from 0.00 to 1.00 were tested. When alpha is close to 1, the hybrid model gives more weight to the Growth-Rate prediction. When alpha is close to 0, the model gives more weight to the Naive lag_1 prediction. The validation results showed that the best alpha value was 0.00.

This result indicates that adding the Growth-Rate prediction into the hybrid formula did not improve the hybrid model performance. Since $\alpha = 0.00$, the final hybrid prediction relied entirely on the Naive lag_1 component. Therefore, the hybrid models were not adopted as the final forecasting model, because the blending process did not provide an actual improvement.

Model	Selected Alpha	MAE	RMSE	MAE%	RMSE%
Growth-Rate XGBoost	—	4,432.11	11,214.91	9.51	23.13
Growth-Rate Random Forest	—	5,211.45	13,300.04	11.18	28.54
Corrected Hybrid XGBoost	0.00	5,969.35	12,928.40	12.81	27.75
Corrected Hybrid Random Forest	0.00	5,969.35	12,928.40	12.81	27.75

Table 5.2 Final forecasting model performance comparison.

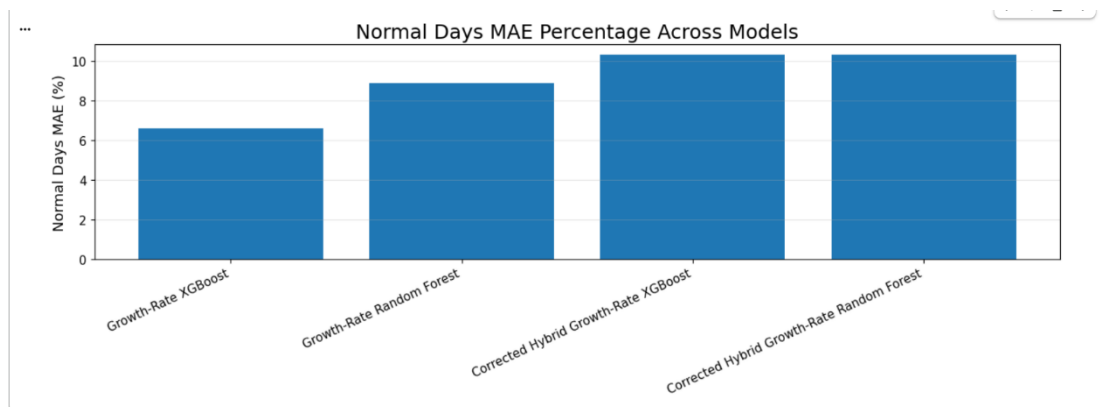


Figure 5.1 Normal days MAE percentage across models.

5.2.3 Final Adopted Forecasting Model

After completing the forecasting experimentation phase and evaluating the different forecasting approaches, the Standalone Growth-Rate XGBoost model was selected as the final adopted forecasting model for the proposed system. The selection was based on forecasting accuracy, error reduction, and overall suitability for integration with the dashboard and recommendation layer.

The final adopted model was compared with Growth-Rate Random Forest. Although the Growth-Rate Random Forest model showed acceptable performance, the Growth-Rate XGBoost model achieved lower forecasting errors and provided stronger stability across the 1446H testing period.

After completing the forecasting experimentation phase and evaluating the different forecasting approaches, the **Standalone Growth-Rate XGBoost** model was selected as the final adopted forecasting model for the proposed system. The selection was based on forecasting accuracy, error reduction, model stability, and overall suitability for integration with the interactive dashboard and recommendation layer.

The final adopted model was compared with the **Growth-Rate Random Forest** model. Although the Growth-Rate Random Forest model showed acceptable performance, the Growth-Rate XGBoost model achieved lower forecasting errors and demonstrated stronger stability across the 1446H testing period.

The final model was trained to predict the weekly growth-rate target, `growth_rate_7`, instead of directly predicting the raw number of Umrah performers. After predicting the expected growth rate, the final predicted Umrah performer count was reconstructed using the `lag_7` value. This approach allowed the model to focus on relative weekly changes and better capture the movement patterns of Umrah performer counts.

Table 5.3: Final adopted model comparison on the 1446H test dataset.

Model	MAE	RMSE	MAE%	MAPE
Growth-Rate XGBoost	4,666.79	11,214.91	9.29%	8.00%
Growth-Rate Random Forest	6,024.08	13,300.04	11.18	10.99%

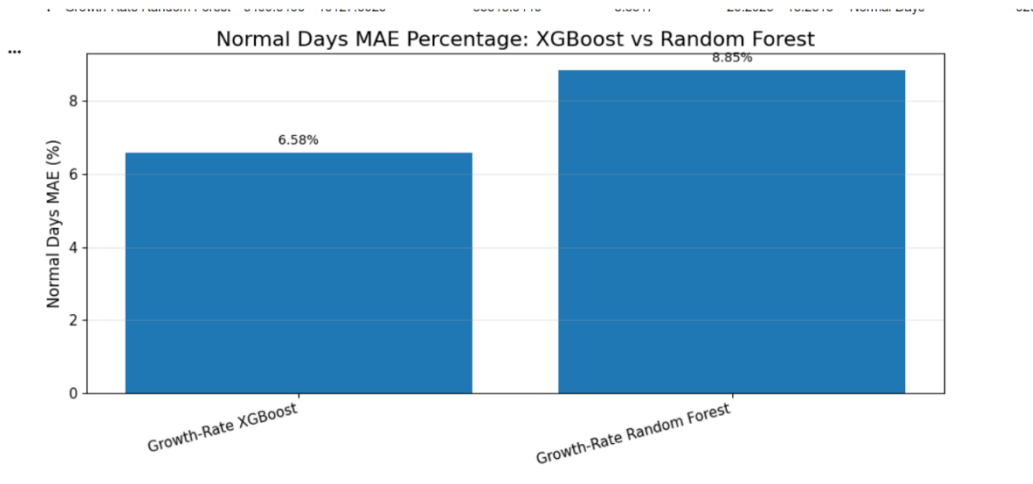


Figure5.2 Normal days MAE percentage comparison between Growth-Rate XGBoost and Growth-Rate Random Forest.

These results show that the Growth-Rate XGBoost model produced the lowest prediction error among the evaluated forecasting models. The MAE value of 4,666.79 indicates that, on average, the model prediction differed from the actual number of Umrah performers by about 4,667 performers. The MAE% value of 9.51% and MAPE value of 8.00% show that the model achieved acceptable forecasting accuracy for the proposed daily crowd-analysis system.

The stronger performance of the Growth-Rate XGBoost model indicates that it was more capable of capturing nonlinear temporal patterns, weekly performer changes, and seasonal fluctuations in the Umrah dataset. Unlike direct forecasting approaches, this model predicted the weekly growth rate and then reconstructed the expected performer count, which helped stabilize the prediction behavior.

5.3 Ramadan and Seasonal Analysis

This section focuses on analyzing model performance during Ramadan, as it represents a high-density seasonal period in the Umrah dataset. During Ramadan, the number of Umrah performers increases significantly, and the daily

variations become rapid and irregular. This makes the forecasting process more challenging compared with normal days.

Several models were evaluated during Ramadan, including LSTM and Prophet. These results were then compared with the updated developed models, namely Growth-Rate XGBoost and Growth-Rate Random Forest. The updated results show that Growth-Rate XGBoost achieved the best Ramadan forecasting performance among the selected models.

Table 5.6: Ramadan Forecasting Error Comparison Across Selected Models

Model	Ramadan MAE	Ramadan MAE%	Interpretation
Growth-Rate XGBoost	23,484.27	17.13%	Best Ramadan performance among the selected models and adopted as the final model.
Growth-Rate Random Forest	28,745.56	20.97%	Acceptable performance, but higher error than Growth-Rate XGBoost.
LSTM	34,592	25.23%	Higher error due to the difficulty of capturing sudden fluctuations during Ramadan.
Prophet	55,297	40.34%	Weakest performance during Ramadan due to limited ability to represent sharp peaks.

Based on the updated comparison, Growth-Rate XGBoost achieved the lowest error during Ramadan, with MAE = 23,484.27 and MAE% = 17.13%. This indicates that the model was more capable of handling the sharp fluctuations and irregular daily changes in Umrah performer numbers during Ramadan compared with the other evaluated models.

The Growth-Rate Random Forest model also achieved acceptable Ramadan performance, recording MAE = 28,745.56 and MAE% = 20.97%. However, its error remained higher than Growth-Rate XGBoost, which suggests that XGBoost was more effective in capturing nonlinear temporal behavior and weekly growth-rate patterns during peak seasonal conditions.

For LSTM, the model recorded a higher error during Ramadan, with MAE = 34,592 and MAE% = 25.23%. Although LSTM is commonly used for time-series forecasting, its performance in this experiment was weaker. This is likely

because the available dataset was not large enough to train a deep learning model effectively for highly volatile Ramadan behavior.

The Prophet model showed the weakest performance during Ramadan, recording $MAE = 55,297$ and $MAE\% = 40.34\%$. This indicates that Prophet was not suitable for representing the sharp increases and strong daily variations in Umrah performer numbers during Ramadan.

Overall, the updated Ramadan results confirm that Growth-Rate XGBoost was the most reliable model during Ramadan. Although Ramadan remained the most difficult period to forecast, Growth-Rate XGBoost provided the best balance between accuracy and seasonal tracking performance.

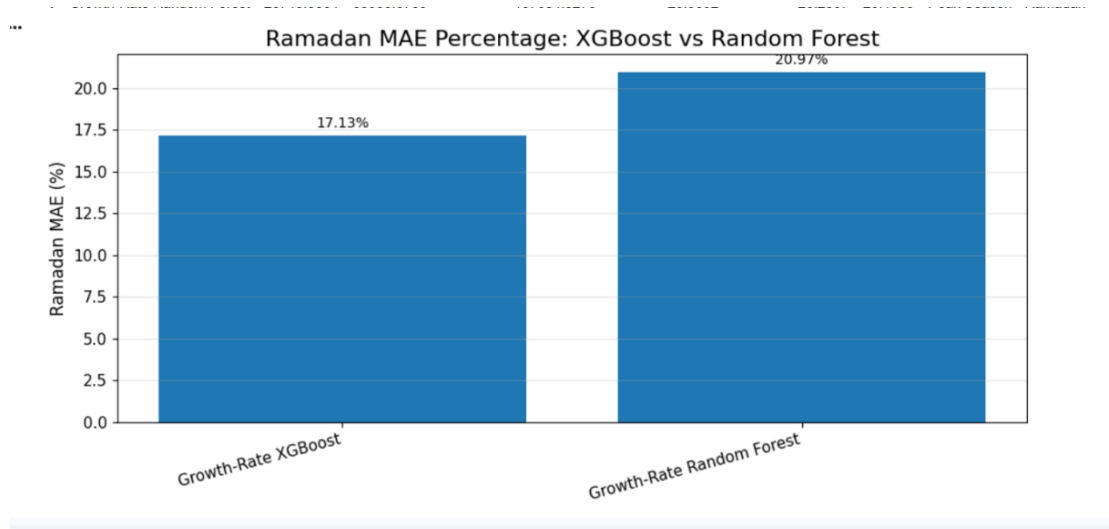


Figure5.3: Ramadan MAE percentage comparison between Growth-Rate XGBoost and Growth-Rate Random Forest.

5.4 Dashboard Results and Recommendation Outputs

This section presents the final dashboard results of the proposed Umrah crowd forecasting system. The dashboard was designed to transform the forecasting model outputs into clear visual indicators and practical recommendation messages. Instead of presenting only numerical predictions, the dashboard guides the user through a simple process that starts with entering

the required information, then displays the forecasting results, crowding level, seven-day forecast visualization, and recommendation outputs.

The dashboard was developed in Arabic to make it more accessible and understandable for Umrah performers. It allows the user to select the required input information and then receive a clear recommendation about whether the selected day is suitable for performing Umrah or whether a less crowded alternative day should be considered.

5.4.1 Main Dashboard Interface

The main dashboard interface represents the entry point of the system. It introduces the purpose of the Umrah crowd forecasting system and provides a clear overview of the main functions available to the user. The interface includes a welcoming message, a short description of the system, and feature cards that summarize the main benefits of the dashboard.

The main page highlights three core functions: smart forecasting, best-day suggestion, and clear recommendation. The smart forecasting feature allows the system to display the expected crowding level. The best-day feature helps suggest a less crowded day when needed. The recommendation feature supports the Umrah performer in making a clearer decision before visiting Al-Masjid Al-Haram.

This page also includes a button that allows the user to start entering the required information. This design makes the dashboard easy to use because the user can understand the system purpose before moving to the prediction process.

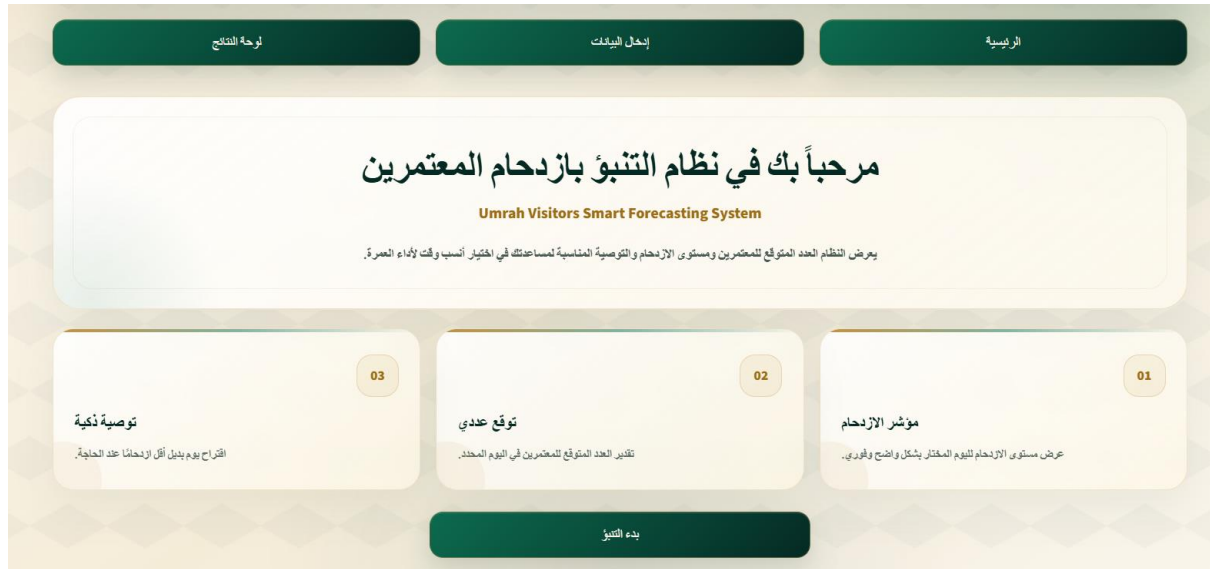


Figure 5.4.1: Home page of the Umrah Visitors Smart Forecasting System dashboard.

5.4.2 User Input Page

The second stage of the dashboard is the input page. In this page, the user enters the required information, including nationality, Hijri month, and Hijri day. These inputs are used to identify the selected date and connect it with the forecasting model and recommendation logic.

The nationality input is included because some periods may have specific access conditions for non-Saudi users, especially around Hajj-related months. The Hijri month and day inputs allow the system to locate the selected date and generate the corresponding prediction result.

5.4.2 Dashboard Input Page

5.4.3 Non-Saudi Restriction During Dhul-Qi'dah

Before displaying the forecasting result, the dashboard applies a validation rule for non-Saudi users during **Dhul-Qi'dah**. If a non-Saudi user selects a date between **1 and 15 Dhul-Qi'dah**, the system displays a restriction message and prevents the forecasting result from being shown.

This rule was added to make the dashboard more realistic and aligned with Hajj-related seasonal access conditions. It also improves the recommendation process because the system does not only generate numerical predictions, but also checks whether the selected date is suitable for the user category before showing the result.

Figure 5.4.3 Non-Saudi Restriction Message During Dhul-Qi'dah

5.4.4 Selected-Day Forecasting Output Cards

After the user submits a valid date, the dashboard displays the main forecasting output using summary cards. These cards include the selected Hijri date, weekday, predicted number of Umrah performers, expected crowding level, and temperature. This layout allows the user to quickly understand the expected condition of the selected day without needing to interpret raw model outputs.

In the shown example, the dashboard displays the selected day, the expected number of Umrah performers, the crowding level, and the temperature. The crowding level is highlighted visually to help the user immediately identify whether the selected day is expected to be low, medium, or high in crowd density.



Figure 5.4.4 Selected-Day Forecasting Output Cards

5.4.5 Final Recommendation Output

Below the output cards, the dashboard displays the final recommendation message. This recommendation converts the model prediction into a direct decision-support message for the user. If the expected crowding level is low, the system indicates that the selected day is suitable for performing Umrah. If the crowding level is medium, the system recommends visiting with caution and choosing a suitable time. If the crowding level is high, the system recommends selecting another day.

In the shown example, the dashboard recommends choosing another day because the expected crowding level is high compared with nearby days. This

makes the dashboard more useful for non-technical users because it provides a clear recommendation instead of only displaying prediction numbers.



Figure 5.4.5 Final Recommendation Output

5.4.6 Seasonal Indicators for Dhul-Qi'dah and Dhul-Hijjah

The dashboard also displays additional seasonal indicators during specific Hijri periods, especially from **Dhul-Qi'dah** to **Dhul-Hijjah**. These indicators are used to provide contextual explanation for Hajj-related crowd effects and Tawaf Ifadah-related crowd pressure when applicable.

During **Dhul-Qi'dah** and from **1 to 9 Dhul-Hijjah**, the dashboard displays the expected number of Hajj pilgrims. This indicator helps explain the possible increase in crowd movement before and during the main Hajj preparation period. Since Dhul-Qi'dah is closely connected to pre-Hajj movement, showing the Hajj-related indicator during this period provides users with a clearer understanding of why crowd levels may increase.

Starting from **10 Dhul-Hijjah until the end of Dhul-Hijjah**, the dashboard displays the Tawaf Ifadah indicator instead of the Hajj count. This is because Tawaf Ifadah is directly related to the post-Hajj ritual period and may contribute to additional crowd pressure around Al-Masjid Al-Haram. Therefore, the dashboard uses this indicator to explain the expected crowd behavior during the days following the main Hajj day.

It is important to note that the Hajj and Tawaf Ifadah indicators are not used as direct training features in the final forecasting model. Instead, they are used in the dashboard as interpretation and recommendation indicators. This means that the forecasting model remains focused on time-series prediction,

while the dashboard provides additional contextual explanation to help users understand the expected crowd conditions.

5.4.7 Seven-Day Forecast Visualization

The dashboard includes a seven-day forecast visualization that starts from the selected date and continues chronologically for the next six days. The plot shows the predicted Umrah performer count for each day and uses color indicators to represent the expected crowding level.

The visualization helps the user compare nearby days and understand how the expected crowd level changes over the short-term forecast window. The colors make the chart easier to interpret, as each point represents the crowding level of a specific day. This comparison supports the recommendation system by helping users identify less crowded alternatives within the upcoming seven-day period.

In the shown example, the seven-day plot shows that some days have high expected crowding, while other days show lower expected crowding. This allows the user to compare the selected day with nearby days before making a decision.



Figure 5.4.7 Seven-Day Umrah Crowd Forecast Visualization

5.4.8 Best-Day Recommendation Component

The dashboard includes an optional best-day recommendation component. This section asks the user whether they want to search for a better day within the nearby seven-day period. When the user presses the best-day button, the system compares the predicted Umrah performer counts within the seven-day forecast window and suggests the day with the lowest expected crowding, excluding the originally selected day.

This component is important because it changes the dashboard from a forecasting-only tool into a decision-support system. Instead of only telling the user that the selected day may be crowded, the dashboard provides a practical alternative that can help the user choose a more comfortable visiting day.



Figure 5.4.8 Optional Best-Day Recommendation Button

5.4.9 Dashboard Result Summary

Overall, the dashboard results demonstrate how the proposed system connects machine learning forecasting with practical decision support. The forecasting model predicts the expected number of Umrah performers, while the

dashboard transforms this output into understandable input-based results, visual cards, crowd-level indicators, seasonal indicators, a seven-day forecast plot, and recommendation messages.

This makes the system useful for Umrah performers because they do not need to interpret raw forecasting values directly. Instead, they receive a clear crowding level, contextual information, and practical guidance that supports their decision-making process before visiting Al-Masjid Al-Haram.

The dashboard therefore supports the main objective of the project by helping Umrah performers understand expected crowd conditions and select a more suitable visiting day based on forecasting results and recommendation logic.

5.5 Discussion

This section discusses the experimental findings obtained from the forecasting models and interprets the reasons behind the observed forecasting behavior. The discussion focuses on explaining why certain forecasting approaches achieved better performance, why some periods such as Ramadan were more difficult to forecast, and how the system results can support crowd analysis and decision support.

In addition to numerical evaluation, this section analyzes the forecasting behavior of the models under different seasonal conditions and discusses the practical implications of the obtained results for crowd analysis and decision support.

5.5.1 Why Growth-Rate XGBoost Performed Better

The experimental results showed that the Standalone Growth-Rate XGBoost model achieved the best overall forecasting performance compared with the baseline forecasting models, hybrid forecasting models, and Growth-Rate

Random Forest approaches. Several factors contributed to the superior performance of this model.

One of the main reasons behind the stronger performance of Growth-Rate XGBoost is its ability to model nonlinear temporal relationships within the forecasting dataset. Umrah performer patterns are highly dynamic and are affected by multiple interacting factors such as seasonality, religious events, weekdays, and previous crowd behavior. XGBoost is well suited for capturing these nonlinear relationships because it builds an ensemble of decision trees that correct prediction errors iteratively.

Another important factor is the use of growth-rate forecasting instead of directly predicting raw performer counts. Rather than forecasting the absolute number of performers, the model was trained to predict the relative weekly change in Umrah behavior. This approach helped the model focus on movement trends and weekly variation, which improved forecasting stability.

The engineered time-series features also contributed significantly to the forecasting improvement. Features such as lag variables, rolling statistical features, difference features, calendar variables, and Ramadan indicators allowed the model to better understand historical Umrah performer behavior and seasonal crowd changes.

In addition, XGBoost demonstrated stronger adaptability during periods with changing crowd intensity compared with Random Forest forecasting models. Although Growth-Rate Random Forest achieved reasonable forecasting performance, XGBoost consistently produced lower forecasting errors and more stable overall results.

Overall, the combination of growth-rate forecasting, engineered temporal features, and nonlinear learning capability allowed the Growth-Rate XGBoost

model to provide the most balanced and reliable forecasting behavior among all evaluated forecasting approaches.

5.5.2 Why Ramadan Was More Difficult to Forecast

The seasonal analysis results showed that forecasting during Ramadan was significantly more difficult compared with normal periods. Although the forecasting model maintained acceptable overall performance, forecasting errors increased noticeably during Ramadan peak days.

One of the primary reasons for this difficulty is the highly volatile nature of Umrah behavior during Ramadan. Unlike normal periods where movement follows relatively stable weekly patterns, Ramadan contains sudden crowd spikes, irregular surges, and rapid changes in daily crowd intensity.

Another important challenge is that the forecasting model represented Ramadan mainly through temporal indicators and seasonal features rather than detailed intra-Ramadan behavioral modeling. While Ramadan-related features helped the model identify the Ramadan season, they could not fully capture all internal variations across different days and nights of Ramadan.

In addition, some external operational factors affecting crowd behavior during Ramadan were not available in the dataset. Examples include real-time operational adjustments, temporary crowd-control policies, transportation conditions, weather variability, and sudden increases in Umrah demand. The absence of these variables made it harder for the model to predict extreme fluctuations.

Despite these challenges, the forecasting system still demonstrated reasonable seasonal tracking capability during Ramadan. The model was able to capture the general upward crowd trend and major seasonal movement behavior, even though the forecasting errors became larger during extreme crowd peaks.

5.5.3 Interpretation of the Hybrid Forecasting Experiments

The hybrid forecasting experiments were conducted to investigate whether combining machine learning forecasting outputs with simpler forecasting approaches could improve prediction accuracy and forecasting stability.

The hybrid forecasting structure combined Growth-Rate forecasting models with a Naive lag_1 forecasting component using an alpha weighting mechanism. Multiple alpha values were experimentally tested during the validation stage to determine the optimal balance between the forecasting components.

However, after correcting the alpha-selection methodology and preventing data leakage during validation, the experimental results showed that the optimal alpha values converged toward 0.00 for both hybrid forecasting models. This finding indicates that the hybrid contribution did not provide meaningful improvement.

In practice, the hybrid models became almost fully dependent on the Naive lag_1 prediction rather than benefiting from the machine learning forecasting layer. This suggests that the hybrid structure introduced additional architectural complexity without producing consistent forecasting gains.

From an academic perspective, the hybrid experimentation phase remained important because it demonstrated that the final model-selection decision was based on experimental evidence rather than arbitrary preference. The results showed that increasing forecasting complexity does not always guarantee better predictive performance.

As a result, the hybrid forecasting approaches were not selected as the final forecasting solution. Instead, the Standalone Growth-Rate XGBoost model was

preferred because it achieved better forecasting accuracy, stronger seasonal stability, simpler architecture, and lower risk of unnecessary overfitting.

5.5.4 Strengths and Weaknesses of the Proposed System

The proposed forecasting system demonstrated several strengths throughout the experimentation and evaluation phases. One of the main strengths is its ability to capture long-term seasonal behavior and temporal Umrah patterns using machine learning forecasting techniques combined with engineered time-series features.

Another important strength is the use of growth-rate forecasting methodology, which improved forecasting stability and allowed the model to adapt more effectively to changing crowd-density conditions. The forecasting system also achieved relatively strong yearly forecasting performance across the 1446H testing period.

The interactive dashboard layer represents another significant strength of the proposed system. Instead of presenting only numerical forecasting outputs, the dashboard transformed the predictions into interpretable visual insights, seasonal comparisons, crowd indicators, and operational analytics that support decision-making.

Despite these strengths, the proposed system also contains several limitations and weaknesses. One limitation is the absence of real-time operational data and live crowd-monitoring integration. The forecasting system currently depends mainly on historical data rather than real-time measurements.

Another limitation is the difficulty of modeling extreme seasonal volatility during Ramadan and peak crowd periods. Although the forecasting model captured general seasonal trends, prediction errors increased during highly dynamic crowd conditions.

The forecasting framework also operates mainly at the daily forecasting level and does not currently support intra-day or hourly crowd forecasting. Therefore, the system provides daily crowd-level insights rather than fine-grained real-time crowd predictions.

Overall, despite these limitations, the proposed system demonstrated that combining machine learning forecasting, feature engineering, and interactive visual analytics can provide a practical framework for Umrah performer forecasting and crowd-level analysis at Al-Masjid Al-Haram.

5.6 Limitations

Although the proposed system achieved strong results in forecasting Umrah performer numbers and supporting crowd-level analysis, several limitations should be considered. These limitations do not reduce the value of the system, but they highlight areas that can be improved in future work.

First, the project faced limitations related to data availability and completeness. Some periods did not have complete daily records, which required preprocessing and estimation to maintain a continuous time-series structure. Although this step was necessary to prepare the dataset and allow the forecasting model to run consistently, it may introduce a degree of uncertainty into the final results. More complete and directly observed daily Umrah data in the future would improve model reliability, reduce estimation uncertainty, and make the recommendation outputs more accurate.

Second, Ramadan was the most difficult period to predict due to highly volatile and rapidly changing Umrah behavior. During Ramadan, the number of Umrah

performers increased sharply and fluctuated strongly from day to day. These rapid changes made the forecasting task more challenging compared with normal days.

Third, the current system does not use real-time data. The forecasting results are mainly based on historical records and available temporal features. The system is not directly connected to live data sources such as cameras, sensors, real-time crowd monitoring systems, or transportation flow data. Therefore, sudden changes that occur on the same day may not be captured immediately.

Fourth, the recommendation system depends on predicted values rather than direct real-time measurements. For example, recommendations such as “Go” or “Choose a better day” are based on the predicted number of Umrah performers and the expected crowding level, not on live crowd density inside Al-Masjid Al-Haram. Therefore, the recommendation outputs may differ from the actual operational crowd situation if unexpected operational changes or sudden crowd increases occur.

Overall, the current system provides a practical and useful framework for forecasting Umrah performer numbers and supporting Umrah-time recommendations. However, future improvements could include adding real-time data, improving Ramadan-specific modeling, expanding the dataset, and developing more detailed hourly forecasting instead of relying only on daily predictions.

5.7 Chapter Summary

This chapter presented the experimental results and discussion of the proposed intelligent forecasting and crowd-analysis system for Al-Masjid Al-Haram. The chapter began by presenting the forecasting results obtained from baseline forecasting models, hybrid forecasting experiments, and machine learning forecasting approaches.

The experimental evaluation showed that the Standalone Growth-Rate XGBoost model achieved the best overall forecasting performance compared with the other evaluated models. The results also demonstrated that hybrid forecasting approaches did not provide meaningful forecasting improvement after correcting the validation methodology and alpha-selection process.

In addition, the chapter presented a detailed seasonal analysis for Ramadan, which represented the most difficult forecasting period due to highly dynamic Umrah behavior and sudden crowd fluctuations. Despite the increased forecasting difficulty during Ramadan, the final adopted forecasting model maintained relatively stable seasonal tracking performance compared with the other evaluated approaches.

The chapter also presented the dashboard and recommendation-system results, where forecasting outputs were transformed into interpretable visual analytics, crowd-level indicators, and Umrah-time recommendations that support operational crowd understanding and decision support.

Furthermore, the discussion section explained the reasons behind the superior performance of the Growth-Rate XGBoost model, the challenges associated with forecasting Ramadan periods, and the interpretation of the hybrid forecasting experiments. The strengths and limitations of the proposed system were also analyzed from both forecasting and operational perspectives.

Chapter 6: Conclusion and Future Work

6.1 Introduction

This chapter presents the conclusion and future directions of the proposed intelligent forecasting and crowd-analysis system for Al-Masjid Al-Haram. It summarizes the main achievements, key findings, limitations, and possible future improvements related to forecasting accuracy, recommendation logic, and dashboard deployment.

6.2 Project Overview and Achievements

This project developed an intelligent system for forecasting Umrah performer numbers and analyzing crowd levels at Al-Masjid Al-Haram. The proposed system combined time-series forecasting, feature engineering, machine learning, dashboard visualization, and recommendation logic into one decision-support framework.

Several forecasting approaches were tested, including baseline models, machine learning models, and hybrid forecasting experiments. Based on the experimental results, the Standalone Growth-Rate XGBoost model was selected as the final adopted model because it achieved stronger forecasting accuracy and more stable performance.

In addition to prediction, the project developed an Arabic interactive dashboard that transforms model outputs into clear visual indicators. The dashboard displays the predicted Umrah performer count, expected crowding level, temperature, seasonal indicators, seven-day forecast visualization, and recommendation outputs. This helps Umrah performers understand expected crowd conditions and choose a more suitable visiting day.

6.3 Main Findings

The results showed that the Standalone Growth-Rate XGBoost model achieved the best overall forecasting performance compared with the other tested approaches. The model was able to capture weekly movement patterns and seasonal changes more effectively than baseline and hybrid models.

The results also showed that Ramadan was the most challenging period to forecast because of rapid daily fluctuations and high crowd density. Although forecasting errors increased during Ramadan, the adopted model still provided the best seasonal tracking performance among the selected models.

Another important finding is that the dashboard and recommendation layer improved the usability of the forecasting results. Instead of showing numerical predictions only, the system converted the outputs into clear crowd levels and practical recommendations for Umrah performers.

6.4 Limitations of the Current Work

Although the proposed system achieved promising forecasting and crowd-analysis results, several limitations remain. First, part of the daily dataset was generated using estimation and weighted-distribution techniques due to the limited availability of complete daily Umrah records. Second, some potentially influential variables, such as real-time crowd movement, entry and exit counts, operational-capacity information, and peak-hour statistics, were not available in the open datasets used in this project. Third, weather information was available only at a monthly level rather than a daily level. Therefore, the forecasting results should be interpreted within the scope and limitations of the available data. Future work may benefit from integrating real-time operational and mobility data to improve forecasting accuracy and recommendation quality.

6.5 Project Conclusion

Overall, the proposed system successfully demonstrated a practical framework for forecasting Umrah performer counts and supporting crowd-level analysis at Al-Masjid Al-Haram. The project showed that combining machine learning forecasting with interactive visualization can make prediction results more useful and understandable for users.

The final Standalone Growth-Rate XGBoost model provided accurate and stable forecasting results, while the dashboard improved the interpretation of these results through visual cards, crowd-level indicators, seasonal explanations, and best-day recommendations.

This project highlights the importance of not only achieving good forecasting accuracy, but also presenting the results in a user-friendly way that supports decision-making. Therefore, the proposed system can help Umrah performers plan their visits more comfortably and make better decisions based on expected crowd conditions.

6.6 Future Work

Although the proposed system achieved promising results, several improvements can be considered in future work.

First, the forecasting model can be improved by using more complete and directly observed daily Umrah data. A larger and richer dataset may help the model better capture seasonal patterns, sudden changes, and irregular crowd behavior.

Second, future versions of the system can integrate real-time operational data from sensors, cameras, transportation systems, or crowd-monitoring platforms. This would allow the system to update predictions dynamically and respond to sudden changes in crowd movement.

Third, the forecasting approach can be extended from daily prediction to hourly or intra-day forecasting. This would provide more detailed crowd information and support better planning during peak periods such as Ramadan and Hajj.

The recommendation system can also be improved by adding personalized recommendations based on user preferences, preferred visiting times, acceptable crowd levels, and user categories. In addition, an AI-based chatbot could be integrated to explain the results and recommendations in a more interactive way.

Finally, the dashboard can be improved by deploying it as a fully hosted web platform connected to live forecasting pipelines. Future versions may also include mobile-friendly design, multilingual support, and advanced interactive analytics.

6.7 Final Remarks

This chapter summarized the final outcomes and future directions of the proposed intelligent forecasting and crowd-analysis system. The project demonstrated that machine learning, time-series feature engineering, dashboard visualization, and recommendation logic can be combined to support Umrah crowd analysis at Al-Masjid Al-Haram.

Future improvements such as real-time data integration, hourly forecasting, personalized recommendations, chatbot support, and live dashboard deployment can further enhance the practical value and scalability of the proposed system.

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Appendix

GitHub Repository:

<https://github.com/Nuhalabdul/intelligent-umrah-crowd-forecasting-system>

Interactive Dashboard Link:

<https://intelligent-umrah-crowd-forecasting-system-bbm9vprvwhhezu7dr5s.streamlit.app/>